



Solar[™]
Decathlon
India



ATRIA
UNIVERSITY

Concept and Low-fidelity Prototyping Report – Feb 2025

Product Division

Team “Gaia”

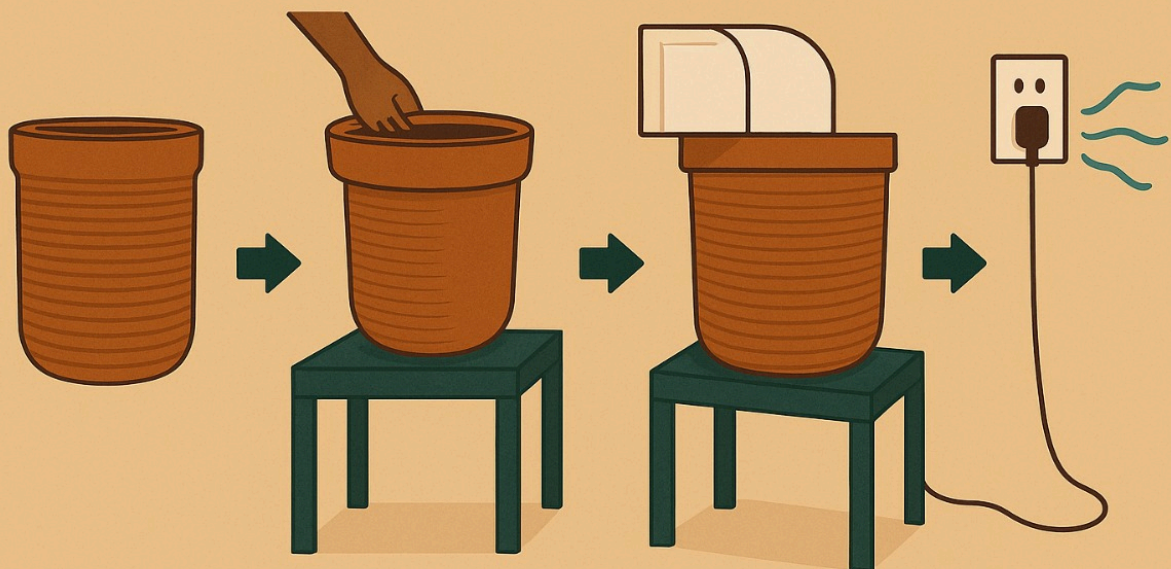
Atria University, Bengaluru



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RESPONSE TO REVIEWERS' COMMENTS

SECTION	REVIEWER'S COMMENT	OUR RESPONSE
REVIEWER 1		
VALUE PROPOSITION	I liked the value proposition. It would have been better to have a slight elaboration on the 'How' part in the value proposition. Having some visual aids for the product ad would have also helped.	visual representation have been added to make the ad more engaging
NOVELTY	The novelty of the solution comes out well. However, certain parameters do not have a like-to-like comparison. For example, water management is reported as per cycle for your product and per day for a cooler. Also, the table mentions 8L/cycle whereas subsequent text mentions 3L/cycle.	Thank you for acknowledging the novelty of our concept. We have corrected the water consumption figure in the Comparative Novelty Analysis table and Innovation Details to clarify it is approximately 3L per day, addressing the previous typo and the comparison unit. The 8L was the capacity of our previous model's tank.
TARGET MARKET	The target market could have been evaluated in much more detail, narrowing down on the geographies of interest and the size of the market within those geographies. Also, the income levels and other demographics could have been used as filters to further arrive at a more accurate target market.	The target market has been more precisely defined by focusing on specific geographies, including major urban centers in hot and humid regions like Mumbai, Chennai, and Bangalore. This geographic focus ensures we're addressing areas with the highest cooling demand. Additionally, demographic factors, such as middle-income households (₹5-30 lakh annual income) and structural constraints like rental apartments and older buildings, have been used to narrow the market. This approach was done with the intent to provide a more accurate and realistic target market, addressing the needs of households seeking cost-effective, retrofittable cooling solutions. Thank you for the valuable suggestions.
USER DESIRABILITY	Liked the findings. I feel that if you had also included people owning ACs to the surveys, you'd have gotten more insights into where the user desirability ends.	Thank you for your feedback. We made sure to include people in our focus group discussions that owned ACs to give a wider range of perspectives essential for coming to a fitting value based pricing. This has been more clearly addressed in the Financial Feasibility section, under the 'User Desirability Testing' subheading.

COOLING PERFORMANCE	The work is as per the expectations. However, there are certain reservations I have w.r.t the cooling performance: 1. The product doesn't seem to adequately cool the entire room. Steps to address that are not mentioned. 2. The humidity is outside the desired band for the product as per the diagram, yet the text says that it is within the comfortable zone. This seems to be a mismatch. If the product cannot cool a room, then the fundamental expectations from the product aren't met. This should be addressed at the earliest.	Thank you for your feedback. We acknowledge your reservations regarding the cooling performance. Regarding whole-room cooling, we recognize the current prototype primarily provides effective localized cooling. We have clarified this limitation and outlined specific planned technical enhancements, focusing on fan/airflow optimization and maximizing terracotta evaporation rates, with the explicit goal of achieving adequate whole-room cooling in future iterations. Concerning the humidity interpretation, we appreciate you pointing out the mismatch. The report's analysis has been revised to accurately reflect that the measured output humidity is at the upper end of standard comfort ranges, noting its potential benefits in dry conditions while acknowledging the need for management strategies (like potential sensor integration or desiccant use) which are under consideration for broader applicability. We agree that cooling a standard room is the fundamental expectation, and the planned future developments are specifically targeted to ensure the product meets this requirement effectively.
CO-BENEFITS	Co-benefits are identified well, and though mostly qualitative, there is some backing for each of the co-benefits. If the presence of algae was not discovered, it shouldn't have been listed, especially when there are other co-benefits with good backing.	Thank you for your valuable feedback. We have carefully revised the co-benefits section to remove the mention of algae, as formal testing did not confirm its presence. We have instead emphasized the other well-supported co-benefits, including natural ambience, thermal benefits, artisan support, and sustainability, all of which are backed by qualitative user interviews and observations.
EASE OF INSTALLATION	The installation guide doesn't mention anything about the tools required, things to keep in mind during installation, water refill process, etc.	Thank you for your feedback, we have updated the installation guide to include key precautions, and a clear, user-friendly water refill process through a top inlet with overflow control. Our solution does not require any additional tools just a simple plug and play action.
TECHNICAL FEASIBILITY	While the feasibility looks good overall certain assumptions do not seem rational. For example, one artisan requiring one week for one pot should be rechecked. If that is indeed the case, the cost of the product should be much higher than what is being reported. Similarly, you've mentioned 'water resistant coatings' to enhance lifespan of terracotta components, without clarity on whether their performance will get impacted.	Thank you for the feedback. We acknowledge the points raised. Regarding pottery production time, while the full process from raw mud to final baking for a single pot takes approximately one week, skilled artisans produce multiple pots simultaneously, ensuring that production quantity is feasible. We are also actively in discussions with local artisans to implement efficiencies that will further reduce production time and maintain our target price point. Regarding water resistant coatings, this treatment is applied to internal electronic components and fans for protection against moisture, ensuring durability and

		functionality without impacting the terracotta's porosity or evaporative cooling performance.
FINANCIAL FEASIBILITY	I think you missed the point here. You were supposed to arrive at a price that people are willing to pay for your product for your value proposition, and that should be backed through some rationale. You have used the survey results and mapped them to your product, without clarity on whether the survey participants were aware of the features and drawbacks of your product. Maybe a separate survey could have been done to discover the value based pricing for your product.	Thank you for your feedback. We acknowledge the concern regarding the clarity of value-based pricing. While a second formal survey was not conducted, we instead hosted a focused user discussion with select participants from the original survey group, where the complete feature set and limitations of TerraTrunk were shared in detail. This focus group helped validate both product relevance and price expectations more accurately. Based on their feedback, we refined our value-based price range to ₹8,000–₹9,000, which better aligns with users' willingness to pay when fully informed of the offering.
GO-TO MARKET STRATEGY	The 4 channels seem to be fair. The industry partner's involvement was discussed only at a technical level. Guidance for go-to-market from the industry partner was missing. Potential assignment of roles/capabilities to existing team members was also missing.	We have now clarified that our industry partner, Ambiator, is primarily involved for their domain expertise and patented technology. For the go-to-market strategy, we are working with industry mentors like Ms. Lakshmi Hiranandani and Mr. Viivek Verma who are experienced in product launches. Additionally, we've updated the proposal to clearly define the roles and contributions of each team member going forward.

SECTION	REVIEWER'S COMMENT	OUR RESPONSE
REVIEWER 2		
VALUE PROPOSITION	The value proposition is explained very well in the document. Try to make the Product Ad more visual, reduce the written content, highlight main features, try to avoid using technical jargon -- IMAC, adaptive comfort etc (which may not be understood by common user).	Thank you for the insightful feedback. We've revised the product ad to make it more visual and user-friendly by reducing text, highlighting key features, and replacing technical jargon with simpler language.
NOVELTY	The concept is novel. The key question is can it generate sufficient cooling capacity to provide thermal comfort in a typical room of 150 sq. ft.	Thank you. Regarding cooling capacity refer to the cooling performance section.
TARGET MARKET	The effort put to survey 50 middle class households is commendable and the results of the survey are useful. A few related queries. What was the typical size of the surveyed houses? Which spaces needed cooling and for how long? What was the view of the households towards conventional evaporative air cooler (which are available in the price range of Rs 5000 -10000)? Are there estimates of	Thank you for your feedback. We've addressed survey findings regarding the typical size of the surveyed houses, the spaces requiring cooling, and the households' views on evaporative air coolers in the updated section.

	potential market (number of households) in Bangalore?	
USER DESIRABILITY	My interpretation is that for assessing user desirability, the potential users are given an opportunity to observe/test the functioning of the product (e.g. test drive of a car). You have reported some interaction with the users during product testing (co-benefit section) and mentioned that the users appreciated earthy petrichor aroma, the calming effect of the white noise from water dripping, and the refreshing humidity, but it is not clear whether they were satisfied with the cooling performance of the system (perhaps such an exercise needs to be done during summer months)	Thank you for your feedback. Taking this into consideration, we hosted a focus group and took their real time feedback to see their opinions with respect to usage and pricing. This has been addressed clearly in the Financial Feasibility section, under ‘User Desirability Testing’ subheading.
COOLING PERFORMANCE	The cooling performance methodology is well designed. It is good to note that both air temperature and humidity are being measured. The results of temperature and humidity are reported in graphs for 60 -90 minutes duration. It seems that the area of the test room is around 50-60 sq ft, and the temperature of the room air decreases by around 0.9 C during the test. It would be interesting to see the results over longer period of time (e.g. 8 - 24 hours duration). If the room is closed, will the humidity continue to rise and effectiveness of the system decreases over time? An important question would be what size of system would be required to provide cooling to a typical room of 120 -150 sq ft? Or this solution is more suited for providing personal thermal comfort to an individual? Another question is regarding the evaporative cooling performance of the pot over longer time duration. In case of earthen pots (used for cooling water) it is observed that the pores (of the clay pot) gets clogged due to salt and impurities present in the water and the cooling performance drops, ultimately resulting in the need to replace the pot. Do you also expect similar behavior? Will you like to specify the specification of water (e.g. TDS in water) to be used to ensure maintaining cooling performance over long periods of time?	Thank you for the positive feedback on our methodology and for your insightful questions regarding performance. We confirm the initial tests presented were for a shorter duration and conducted in a compact room (approx. 55 sq ft), contributing to the observed localized cooling effect. We have now included 24-hour performance data to better illustrate trends, and longer-duration empirical testing is planned. Your point about potential humidity buildup in a closed room over time is well-taken. We are exploring mitigation strategies, including recommending periodic ventilation and investigating humidity sensor integration to potentially control water flow by adding threshold. Regarding application, the current system performs well as a personal/spot cooler. Achieving efficient cooling for a typical 120-150 sq ft room is our development goal, requiring the planned optimizations to fan systems and the evaporative matrix. Long-term performance and potential pore clogging are critical considerations. Our approach includes integrating cellulose fiber filters to reduce impurities, and half the use of terracotta beads ensuring easy replacement of the modular terracotta beads, and investigating protective coatings to maintain performance. We will also consider providing water quality recommendations (like TDS

		levels) for users.
CO-BENEFITS	You have done a good job of listing the co-benefits.	Thank you for your encouraging feedback.
EASE OF INSTALLATION	The sketches are good to show the product in a room setting. May like to add the quantity and method of filling the water in the pot. Also, the excess water is likely to drip from the outer surface. Is there a need for providing a tray to collect this water?	Thank you for your kind words. Details about the quantity of water have been added in the novelty section, and the method of filling water is now shown with clear visual representations. Our solution also includes a marked maximum water level to prevent overflow, eliminating the need for a tray. q
TECHNICAL FEASIBILITY	It is a plug and play system so only requires sufficient space and electricity supply.	
FINANCIAL FEASIBILITY	Very well explained.	Thank you for the feedback and appreciation.
GO-TO MARKET STRATEGY	Well thought and well articulated Go-To Market Strategy. Choice of Ambiator as industry partner is excellent.	Thank you for the appreciation. We're glad the choice of Ambiator as our industry partner aligns with your expectations. Since our target market differs from Ambiator's, we are complementing their technical input with go-to-market mentorship from professionals who specialize in market strategy.
ADDITIONAL COMMENTS	The product looks promising. Overall, the team must be complemented for good quality focused work on product development and good quality documentation covering all aspects. The work is well-supported by an appropriate industry partner. The key question is - Can the product deliver adequate cooling to a typical room of 120 -150 sq ft in a residential setting? Looking forward to next stage of product development.	Regarding the key question: Our current prototype demonstrates effective localized/spot cooling in smaller test spaces (50-60 sq ft). Achieving adequate cooling for a typical 120-150 sq ft room is the primary technical challenge and the explicit goal for our next stage of development, focusing on optimized airflow and enhanced evaporation as outlined in our report. We look forward to validating this capability in future iterations.

TEAM SUMMARY

Product Name: TerraTrunk

Team name: Gaia

Institution name: Atria University

Competition division: Product division

Team Member	Education	Skills	Contribution to the following contests
Jewel M Jain	B.Tech Digital Transformation	Financial Analysis, Business Development, Strategy	Value Proposition, Financial Feasibility Go-to-Market Strategy
Pruthvi Naresh	B.Tech Digital Transformation	Marketing and Communication, Market Analysis	Target Market, Go-to-Market Strategy, User Desirability
Gaurav Shahare	B.Design Energy Sciences	User Experience, Product Development, Marketing	User Desirability, Target Market, Cooling Performance
Manoj Pal	B.Tech Interactive technologies	Research and Innovation, Product Development	Co-benefits Ease of Installation Technical Feasibility
Jimil Doshi	B.Tech E-mobility	Research, Market Analysis	Target Market Financial Feasibility Go-to-Market Strategy
Arnav Patil	B.Tech Interactive Technology	Engineering, Research, Marketing and Communication	Cooling Performance Co-benefits Technical Feasibility
Laxman Prajapati	B.Design Industrial Product Design	Product Prototyping, Design Thinking, Sketching	User Desirability, Prototype Ease of Installation Cooling Performance
Marwa Nazeem	BBA Digital Transformation	Business Lead, Financial Analysis, Market Research	Market Analysis, Value Proposition, Go-to-Market Strategy, Financial Feasibility
Deepak Grover	B.Tech Digital Transformation	Engineering, Product Development, Research	Ease of Installation Technical Feasibility Co-benefits, Documentation
Roshan Bharti	B.Design Design in Interactive Technologies	User Experience, Product Development, Market Analysis	User Desirability Cooling Performance Ease of Installation
Kethan Dande	B.tech E - Mobility	Market Research, Procurement, Electronics and Battery.	Cooling performances, Co-benefits, Ideation
Sarath Babu	B.Tech Interactive Technology	Technical Analysis, Product Development	Development, Technical Feasibility

Background of the lead institution:

Atria University offers a range of **interdisciplinary degree programs**, including Digital Transformation (DT), Interactive Technologies, Life Sciences, Mobility, and Energy Sciences, Design. The DT program combines business and data science, making it relevant for the financial and strategic aspects of the challenge. The Interactive Technologies program covers fields like AR/VR, IoT, and electronics, which are essential for innovative cooling solutions. These courses, along with the university's focus on project-based learning, prepare students for contests like cooling performance and co-benefits.

Resources:

Below there is a basic list of equipment and tools that the team has access to

Lab:

IoT Lab Equipment	Fabrication Lab Equipment
Microcontrollers (Arduino, ESP32, Raspberry Pi)	3D Printer
Sensors (temperature, humidity, pressure, proximity, etc.)	Materials (filaments, PCBs, wood, metals, acrylics)
Actuators (motors, relays, solenoids) and RFID/NFC modules	Adhesives and tapes
Wireless Communication Modules (Wi-Fi, Bluetooth, Zigbee)	Soldering Machine
Breadboards, Jumper Wires, Resistors, Capacitors	Hand Tools (pliers, screwdrivers, cutters, etc.)
Power Supplies and Batteries and Multimeters	Drilling Machines
IoT Development Kits and Ethernet Modules	Power Tools (dremels, grinders, etc.)

**Faculty Lead:**

Jithin CV is an Assistant Professor at Atria University with over 8 years of experience in training and product development, specializing in robotics, embedded systems, and STEM education.

INDUSTRY PARTNERS

Ambiator Private Limited is a climate-appropriate cooling technology firm that is solving for human thermal comfort. The first product that is launched is the Ambiator 5TR which is a modular, scalable cooling solution for Residential, Commercial, Institutional and Industrial use.

<https://www.ambiator.com/>

USERS & STAKEHOLDERS:

Our primary users are **low to middle-class residents** in urban and semi-urban areas, who struggle with rising indoor temperatures but cannot afford traditional air conditioning solutions. Key stakeholders include:

Stakeholders	Key Partners	Role
Terracotta Artisans and Manufacturers	Primary users of our cooling retrofit solution.	Surveys, social media discussions, and pilot installations.
Architects & Designers	Key influencers in retrofitting solutions for existing buildings.	Design workshops, feedback sessions.
HVAC Consultants	Experts in ventilation and energy-efficient cooling.	Technical discussions, industry events.
Product Vendors	Provide components for the cooling system.	Supplier partnerships and testing facilities.
Investors & Partners	Help with funding and scaling production.	Networking events, sponsorships.
Government & NGOs	Potential supporters for sustainable cooling initiatives.	Policy discussions, collaboration on affordability programs.

WHY?

Context & Problem

Indian summers are becoming **hotter and longer**, making indoor cooling essential. However, **air conditioners are expensive**, both in **initial cost (CapEx)** and **long-term electricity consumption (OpEx)**. This makes them **unaffordable for low and middle-class households**, who often resort to inefficient and unsustainable cooling methods.

Furthermore, **existing residential buildings** lack efficient cooling infrastructure, **increasing energy consumption, discomfort, and carbon emissions**.

Opportunities

- **High demand for affordable cooling solutions:** Millions of low/middle-income households need cost-effective alternatives.
- **Retrofit potential:** Instead of installing new cooling systems, our solution can **improve existing buildings**.
- **Climate goals:** Energy-efficient cooling aligns with India's **sustainability and carbon reduction commitments**.

Challenges

- **Affordability:** Designing a low-cost alternative that performs effectively.
- **Space Constraints:** Many low-income homes have **limited installation space**.
- **Scalability:** Ensuring ease of installation across different building types.

WHAT?

Our solution is a **cost-effective retrofit cooling system** designed for **low to middle-income households**. It integrates: Passive cooling techniques to reduce heat gain.

Energy-efficient active cooling for enhanced comfort.

Affordable & scalable design to suit different residential buildings.

System & Components

COMPONENTS

S.no	Items	No. of items used
1	Teracotta Pot	1
2	Sunboard	1
3	Fans	2
4	Water Pump	1
5	Teracotta Beads	100
6	Charging Adaptor	1

Designed for **easy installation** in existing homes.

Uses **minimal electricity compared** to traditional ACs.

Limitations

- May not work effectively in **extremely high-humidity regions**.
- Cooling effectiveness depends on **building orientation and ventilation**.

Objectives & Performance Metrics

Objective	Assessment Metric
Reduce indoor temperature	Measured cooling efficiency (°C drop)
Lower energy consumption	kWh savings vs. AC units
Improve affordability	Cost per unit vs. traditional cooling solutions
Ease of installation	Time and effort required for retrofit

HOW?

Product Development Process

1. Comparative Analysis of Existing Solutions:

- Studied Humidifiers, Window Ventilation Systems, Air Conditioners (ACs), and Smart Thermostats.
- Identified both successful features (energy efficiency, comfort, accessibility) and unsuccessful features (high energy consumption, maintenance issues, limited effectiveness).
- Highlighted specific technical data (e.g., humidifier wattage, MEV consumption, AC wattage).

2. Detailed Evaluation of Each Solution:

- For each cooling method, discussed real-world advantages and limitations.
- Provided technical performance figures (energy consumption, maintenance challenges).
- Connected technology features (like Smart Thermostats) to user experience (remote access, smart home integration).

3. Secondary Market Research:

- Analyzed secondary datasets (AC prices, fan vs. AC count across companies in India).

- Summarized temperature and pressure data trends over time.
- Highlighted affordability issues (many companies have more fans than ACs) reflecting market gaps.

4. Primary User Research:

- Conducted interviews and surveys with **50 participants**, aged 20–40 years.
- Captured both qualitative and quantitative insights (some interviews filmed, some by phone).
- Supported findings with external sources like Statista confirming that 25–34-year-olds are the largest spenders on cooling appliances in India.

5. Mixed-Method Research Approach:

- Combined **primary research** (interviews, surveys) with **secondary data** (market reports, online datasets).
- Ensured broader validation by cross-referencing participant opinions with market data trends.

Conceptualization & Prototyping

- Developed multiple design iterations.
- Tested feasibility of **passive + active cooling hybrid models**.

Performance Testing & Refinement

- Evaluated cooling effectiveness in

real-world conditions.

- Optimized for **low-cost materials and modularity**.

Pilot Deployment & Feedback

- Installed prototypes in target communities.
- Gathered user insights for further refinements.

Next Steps

- Scaling up pilot installations.
- Optimizing materials for **lower cost & higher efficiency**.
- Partnering with **housing communities & local manufacturers**.

CONTEST AREA

Introduction

In developing our product solution, we have meticulously addressed all ten contests of the Product Division, ensuring a comprehensive approach that meets the diverse expectations of a successful product. Our solution integrates key aspects from each contest, demonstrating a well-rounded and thoughtful design. By considering all critical areas, we have crafted a product that aligns with the highest standards of innovation, functionality, and impact.

VALUE PROPOSITION

TeraTrunk – Natural Cooling, Reimagined

A Sustainable Solution for Affordable Urban Cooling

Why sweat through summer when nature has the answer?

As India faces rising temperatures due to the Urban Heat Island (UHI) effect, the demand for affordable, energy-efficient cooling has never been greater. However, traditional air conditioning systems are expensive, and energy-intensive, and contribute significantly to carbon emissions and urban heat emissions.

TeraTrunk is an innovative, terracotta-based cooling system that merges ancient wisdom with modern technology to deliver effective cooling while minimizing environmental impact and energy costs.

Unlike conventional cooling systems, TeraTrunk leverages natural evaporative cooling through its porous terracotta structure, enhanced by a dual-fan airflow system, to reduce room temperatures by 4-6°C. Operating at just 50W of power—90% less than an air conditioner—TeraTrunk provides a cost-effective and eco-friendly solution tailored for urban middle-class households.

Why Terra Trunk?

- **Affordable Cooling:** Priced at ₹5000 - ₹6,000, TeraTrunk offers an economical alternative to ACs, with annual operating costs under ₹200.
- **Eco-Friendly Performance:** Zero refrigerants, biodegradable materials, and low electricity consumption reduce carbon footprint.
- **Portable & Easy Setup:** Plug & play installation in under 15 minutes—no permanent modifications required.
- **Energy Savings & Comfort:** Reduces household electricity bills by over 90%.

- maintaining optimal humidity levels (40-60%) for better indoor air quality.
- Sustainable Impact: Helps mitigate urban heat emissions and lowers CO₂ output from traditional cooling methods.

Environmental and Market Impact

If just **1% of Bangalore's households** (10,000 units) adopt TeraTrunk, we can:

- Reduce CO₂ emissions by over 2,500 metric tons annually—equivalent to taking 500 gasoline-powered cars off the road each year.
- Cut household electricity bills by at least 30% in peak summer months.
- Reduce urban heat emissions, helping mitigate the UHI effect in densely populated areas.

Target Market & Growth Potential

India's residential cooling industry is projected to grow at a CAGR of 17.58% (2025-2030), reaching USD 10-15 billion by 2030. However, 86% of Indian households still rely on ceiling fans due to the high cost of air conditioning. India Commercial Air Conditioner Market Forecast Report 2025-2033

TeraTrunk fills this market gap, offering an affordable, scalable, and environmentally friendly cooling alternative that aligns with India's energy efficiency goals.

Market Opportunity & Scalability

- Middle-class urban households (₹40,000 - ₹80,000 income bracket) represent 118 million potential customers in India alone.
- Strong demand in hot metro regions such as Bangalore, Mumbai, Chennai, Delhi NCR, and Hyderabad.

- High adoption potential due to low cost, ease of use, and sustainability benefits.

Investor Elevator Pitch

The future of sustainable cooling lies in **affordability, energy efficiency, and low environmental impact**. TeraTrunk is a **cost-effective, electricity-saving solution** that **addresses India's urban heat crisis** by providing a natural cooling system for the 86% of households that cannot afford air conditioning.

TeraTrunk operates at just **50W**, compared to **1,500W for traditional ACs, reducing electricity costs by over 90%**. By leveraging terracotta's natural evaporative cooling properties, **TerraTrunk is projected to lower room temperatures by 4-6°C**, providing relief to urban middle-class households struggling with extreme summer heat.

With urban heat islands intensifying temperatures by 4-7°C in Indian cities and cooling energy demand projected to increase by 160% under climate change scenarios, TeraTrunk is positioned for exponential growth.

Our target market includes middle-class urban households with a combined market size of over 118 million potential buyers. With a competitive price point of **₹3,500 - ₹5,000**, we have achieved 40% gross margins while delivering real environmental impact.

With scalable manufacturing through partnerships with traditional artisan clusters, we are seeking investment to refine production, scale distribution, and expand across India's hottest urban centers.

Join us in revolutionizing home cooling with TeraTrunk.

Why Invest in TeraTrunk?

 **Eco-Friendly & Sustainable** – Uses recyclable materials, requires no refrigerants, and lowers CO₂ emissions.

 **Mass Adoption Potential** – 90% of Indian households use ceiling fans, showcasing a market shift opportunity.

 **Scalability & Expansion** – Portable, easy-to-install, adaptable to various environments, and scalable across multiple urban markets.

 **High ROI for Investors** – 40% gross margin with a fast-growing target market projected to exceed USD 10 billion by 2030.

With a strong market demand, a scalable business model, and significant environmental benefits, **TeraTrunk represents a unique and profitable investment opportunity in the future of sustainable cooling.**

Conclusion: The Future of Affordable, Sustainable Cooling

TeraTrunk is not just a product—it’s a movement towards a cooler, more sustainable future. By merging traditional cooling methods with modern engineering, we bridge the gap between inefficient fans and costly ACs, making sustainable cooling accessible to millions.

- Affordable for middle-class urban families struggling with high cooling costs.
- Scalable across Tier 1 & Tier 2 cities, tapping into a massive untapped market.
- Eco-friendly alternative that aligns with India’s climate action goals.

TeraTrunk: Feel the difference. Save the planet.  

NOVELTY

Expected Results

This section presents a comparative analysis of our Terracotta-based Passive and Active Cooling System with two commonly used cooling alternatives: Air Conditioners and Conventional Air Coolers. We aim to highlight the unique features and innovations that distinguish our solution in the residential cooling market.

Comparative Novelty Analysis

Novelty Parameter	Our Terracotta Cooling System	Air Conditioners	Conventional Air Coolers
Cooling Technique	Dual cooling mechanism combining traditional evaporative cooling through terracotta material with engineered convection cooling enhanced by dual 12V GPU fans	Compression-based cooling using refrigerants and heat exchange between indoor and outdoor units	Simple evaporative cooling using water and cooling pads with single-directional airflow

Energy Consumption	Ultra-efficient operation at less than 50W total power consumption	High energy consumption (1000-2000W) requiring significant electrical infrastructure	Moderate energy consumption (150-250W)
Environmental Impact	Zero refrigerants, natural materials, and minimal carbon footprint	Uses harmful refrigerants with high Global Warming Potential (GWP)	Low direct environmental impact but higher energy consumption than our solution
Thermal Efficiency	4-6°C temperature reduction using passive and active cooling technologies working in tandem	10-15°C temperature reduction but with proportionally higher energy consumption	2-4°C temperature reduction with limitations in humid conditions
Water Management	Closed-loop water circulation system with minimal water consumption (3L/day) and optimized evaporation	No water used in cooling process	High water consumption with open system (10-15L/day)
Portability	Fully portable design with integrated wheels, lightweight construction (10kg dry weight), and no installation requirements	Fixed installation requiring professional setup and permanent modifications	Semi-portable with water tank limitations and bulky design

Innovation Details

1. Unique Hybrid Cooling Mechanism

Our system introduces a reimagined approach to residential cooling by integrating:

Terracotta Body: The specially designed 3-foot terracotta structure features a central divide that creates optimized airflow paths, maximizing the evaporative cooling effect while maintaining structural integrity.

Dual Fan Configuration: Unlike conventional single-direction cooling systems, our dual 12V GPU fan arrangement creates a balanced airflow pattern that:

- Draws ambient air through the water-saturated terracotta surface
- Accelerates natural convection currents
- Distributes cooled air uniformly throughout the space

Enhanced Thermal Exchange: The terracotta's microporous structure provides significantly greater surface area for evaporative cooling compared to conventional cooling pads, resulting in more efficient heat transfer without chemical refrigerants.

2. Revolutionary Energy Efficiency

Our system achieves unprecedented energy efficiency through:

Ultra-Low Power Consumption:

Operating at less than 50W total, our system uses approximately 95% less electricity than conventional air conditioners and 70% less than standard air coolers.

Passive Cooling Amplification: The terracotta material continues to provide passive cooling even during power interruptions, making the system resilient against power outages.

Optimized Component Selection: Each electrical component has been specifically selected for maximum efficiency, with the dual fans and water pump operating at 12V for ideal performance-to-consumption ratio.

3. Innovative Water Management System

Unlike conventional air coolers that require frequent refilling and create humidity issues, our system features:

Closed-Loop Water Circulation: A precision 12V water pump distributes water evenly across the terracotta surface, eliminating water waste and maintaining optimal moisture levels.

Water Conservation Design: The system consumes approximately 3L of water per day, with minimal evaporative loss compared to traditional coolers.

Balanced Humidity Control: The terracotta material naturally regulates humidity, maintaining optimal levels between 40-60% without creating the excessive moisture common with conventional evaporative coolers.

4. Forward-Looking Modular Design

Our system has been engineered with adaptability and future enhancement in mind:

Scalable Architecture: The core design principles can be applied to create units of

various sizes, from personal cooling devices to larger installations for commercial spaces.

Technology Integration Readiness: The system infrastructure is prepared for future enhancements including:

- IoT connectivity for remote operation and monitoring
- Smart sensors for automated temperature and humidity control
- Integration with home automation systems
- Solar power compatibility for off-grid operation

Customizable Configuration: The modular components allow for adaptation to different room layouts and cooling requirements without requiring permanent installation or modifications.

5. Cultural Heritage With Modern Engineering

Our innovation uniquely bridges traditional cooling methods with contemporary technology:

Revival of Traditional Knowledge: The system leverages centuries-old principles of terracotta cooling that have been used across cultures, particularly in hot, arid regions.

Modern Engineering Enhancement: We've applied computational fluid dynamics and thermal modeling to optimize the traditional approach, creating a solution that outperforms its historical predecessors.

Artisanal Manufacturing Potential: The system creates opportunities for traditional pottery communities to participate in modern, sustainable technology production, creating socioeconomic benefits beyond the direct environmental impact.

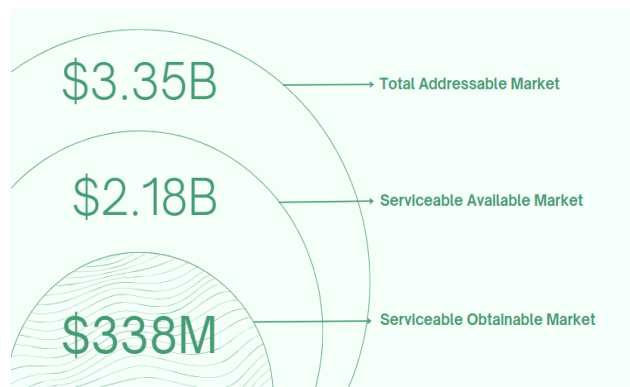
Market Differentiation

Our Terracotta-based Passive and Active Cooling System stands as a truly novel alternative in the residential cooling market, addressing the critical gap between ineffective

fans and inaccessible air conditioners. By combining the best aspects of traditional cooling methods with precision engineering and modern technology, we've created a

solution that is not merely an alternative to existing products but a new category altogether—sustainable cooling that doesn't compromise on performance, affordability, or environmental responsibility.

TARGET MARKET



Total Addressable Market

India's total addressable market for residential cooling includes the entirety of its household base—approximately 330 million households as of 2021¹. Cooling is already a critical need across the country, with ceiling fans present in over 90% of homes, making them the most widely used cooling device². However, the penetration of air conditioners remains low, reaching only 6–8% of households, primarily due to affordability issues and infrastructural limitations. This discrepancy highlights a significant “cooling gap,” representing over 280 million households that desire better thermal comfort but currently rely on insufficient means. The market value of air conditioning alone was estimated at ₹29,500 crore (~\$3.35 billion) in 2022, with a projected CAGR of 17.58% from 2023 to 2028, reflecting rapidly growing demand³. As energy needs increase, this segment has profound implications for both consumer comfort and sustainability, with cooling already consuming 40% of residential

electricity, expected to rise to 60% by 2030 if efficient alternatives are not adopted.

Serviceable Available Market

Within the total market, the serviceable available market includes households located in hot and humid climate zones, where the need for cooling is both urgent and sustained across much of the year. This includes major urban regions like Mumbai, Chennai, Bangalore, Hyderabad, and parts of Delhi NCR, which fall under India's Tropical Wet and Subtropical Humid zones. These geographies not only experience higher average temperatures and humidity but also have dense urban populations that intensify the urban heat island effect. Based on national climate and census data, approximately 215 million households are situated in such regions, forming a large portion of the country's unmet cooling demand. These areas also align with significant residential construction activity and population growth, increasing both demand and installability potential for modern cooling solutions. Considering that the total residential cooling market is valued at ₹29,500 crore (\$3.35 billion), and assuming this subset represents about 65% of that demand (215 households out of 330 million), the SAM can be valued at approximately ₹19,175 crore (\$2.18 billion). This estimate reflects the concentration of unmet cooling needs in climate-stressed regions where traditional air conditioning penetration is low, but thermal comfort is critical.

Serviceable Obtainable Market

The Serviceable Obtainable Market narrows the opportunity down to a more specific segment: urban middle-income households in hot and humid regions that face structural, financial, or environmental constraints to installing traditional air conditioners. This includes households in rental apartments where

¹ <https://www.ice360.in/app/uploads/2021/06/ICE-360-Proposal.pdf>

² <https://www.clasp.ngo/wp-content/uploads/2024/01/CLASP-2021-Annual-Report.pdf>

³

modifications are restricted, older buildings with outdated electrical infrastructure, and eco-conscious developments that prioritize sustainable energy use. According to ICE360 (2022), approximately 31% of India's population falls within the middle-income group (₹5–30 lakh annual income), with a large share concentrated in urban Tier 1 and Tier 2 cities. The estimated SOM comprises 50–70 million households that are underserved and actively seeking cost-effective, plug-and-play, or retrofittable cooling solutions. This segment represents the most immediate and realistic customer base for our innovative residential cooling retrofit solution. that 15–20% of the SAM (₹19,175 crore) falls under this category⁴, the monetary value of the SOM is estimated to be between ₹2,876 crore and ₹3,835 crore (~\$338 million to \$453 million). This subset represents the most realistic early adopters and pilot customers for scalable deployment.

Survey Insights & Market Potential | Bengaluru

As part of our market research, we surveyed 50 middle-income households in Bangalore to understand their cooling needs and preferences. The typical size of the surveyed homes ranged from 600–1,200 square feet, which aligns with the typical middle-income household in the city.

The survey revealed that the primary spaces requiring cooling were the living rooms and bedrooms, with these areas needing cooling for 4–6 hours per day, especially during the hotter months. Regarding conventional evaporative air coolers, priced between ₹10,000 to ₹15,000, the feedback was mixed. While 12% of respondents found them cost-effective, the majority expressed dissatisfaction. Key issues included insufficient cooling in humid conditions and excessive noise, which made them less suitable for extended use.

4

https://iforest.global/2021/11/how-are-indian-households-cooling-themselves/?utm_source=chatgpt.com

USER DESIRABILITY

Primary Research

Techniques Used

To assess the desirability of our cooling retrofit product, we conducted on-ground research by visiting homeowners in Bangalore. Our research methods included surveys, interviews, and observations to understand user preferences regarding cooling solutions. The focus was on evaluating user-friendliness, affordability, and overall satisfaction with potential alternatives to air conditioning.

Surveys & Interviews

We conducted face-to-face surveys with **50** homeowners in Bangalore, asking **20** structured questions about their cooling needs, budget constraints, and product expectations.

Survey Details:

- **Sample Size:** 50 respondents
- **Questionnaire:** 20 questions
- **Target Group:** Middle-class homeowners who do not own an air conditioner or are looking for alternatives.

Key questions included:

- What cooling methods do you currently use?
- What are the biggest challenges with existing solutions?
- Would you prefer a product that offers better cooling than a fan but is more affordable than an AC?
- How much are you willing to spend on an alternative cooling solution?

Observations

During the surveys, we observed user behavior and documented key insights

1. **High Initial Cost of ACs** – Most respondents found air conditioners too expensive to purchase and operate.
2. **Need for an Intermediate Solution** – Many users expressed interest in a product that cools better than a fan but is cheaper than an AC.
3. **Affordability is Key** – Users stated they cannot afford more than ₹5000 for a cooling device.
4. **Preference for Compact & Portable Design** – Homeowners preferred a small, movable cooling device rather than a fixed installation.

Quantitative and Qualitative Results of User Testing

Quantitative Findings (Survey Data & Metrics) Our data revealed the following user preferences:

Factor	Survey Result
Willing to try a cooling solution better than a fan but cheaper than an AC	88% said yes
Cannot afford more than ₹5000 for a cooling device	76% agreed
Prefer a compact and portable cooling solution	84% preferred it
Concerned about the high cost of ACs	91% mentioned this

Qualitative Findings (User Testimonials & Case Studies)

♦ Home User Testimonial

"I don't want to buy an AC because of the high cost, but a fan alone isn't enough in summer. If there's something in between, I'd love to try it." – Ramesh K., Bangalore

♦ Case Study: Budget-Conscious Homeowner

A Bangalore-based family living in a 2BHK apartment shared their experience:

- ✓ Avoided buying an AC due to high upfront cost and electricity bills.
- ✓ Looking for a cost-effective cooling option below ₹5000.
- ✓ Preferred something that does not require installation and is portable.

Supporting Documentation

- Images of survey interactions with homeowners.
- Screenshots of survey responses.
- Photographs of existing cooling setups in homes.

Note: Refer to 'Financial Feasibility' Section, 'User Desirability Testing' subheading for further information on how user desirability has evolved over the course of product development

COOLING PERFORMANCE

1. Introduction

This report evaluates the cooling performance of the TerraCool system, aimed at reducing cooling energy while maintaining adaptive comfort. The testing follows the "Cooling Performance Contest" protocol, comparing the prototype's cooling effect against a baseline case to validate the evaporative cooling principle and assess its potential for effective room cooling.

2. Testing Procedure and Iterations:

Testing was conducted using a modified Approach A protocol, utilizing two rooms of identical dimensions within a three-story building complex in Bangalore. One room, located on the second floor, served as the control room. The other room, located directly below the first floor, served as the test room. The rationale for this setup was to leverage the potentially higher ambient temperatures on the upper floor due to increased solar radiation while acknowledging that the sandwiched position of the floors would result in relatively similar thermal conditions.

Rooms Description:

- **Dimensions:** Both the control and test rooms had dimensions of [[1.8m x2.2m x 2m].
- **Type of Room:** Both rooms were working single rooms in a residential setting.
- **Windows/Doors:** One window was present in each room and was kept closed during testing.

Sensor Placement:

- **Control Room:** One DHT-11 sensor connected to an ESP32 was placed in the control room to monitor ambient conditions.
- **Test Room:** Three DHT-11 sensors connected to ESP32s were used:
 - a. One sensor was placed near the intake area of the TerraCool system (Chamber 1).
 - b. One sensor was placed near the output area of the TerraCool system (Chamber 2, near the fan).
 - c. One sensor was placed 1 meter away from the TerraCool system's output.
- All sensors were positioned to avoid direct exposure to the airflow from the fan, minimizing potential measurement errors due to forced convection.

Iteration 1:

A preliminary test was conducted using an initial prototype design. Four DHT11 sensors were used. One was in the Control Room, one in the middle of the Test Room, and two sensors were near the fans. The Data for temperature was taken.

Baseline Test: A Baseline test without the product was conducted to check the effect of temperature only with the fan. The baseline test includes Pedestal Fan.

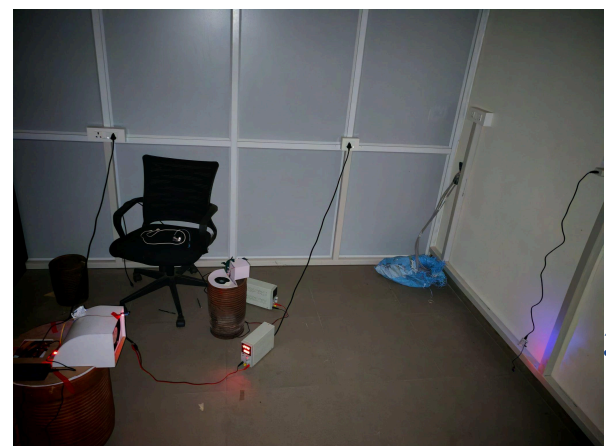
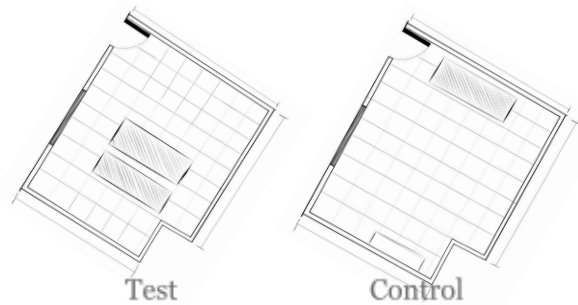
Iteration 2 (Final Prototype):

The finalized TerraCool prototype, as described in Section 2.1, was tested.

Data was collected, focusing on temperature and humidity changes.

Baseline Test: A Baseline test without the product was conducted to check the effect of temperature only with a fan. The baseline test includes Pedestal Fan.

Data Logging: Temperature and humidity data were continuously logged using the ESP32 microcontrollers and transmitted to a cloud-based system for all tests.



3.

Key Observations and Iterations

This section describes the observations and learnings from each testing iteration, justifying the design changes.

Initial Testing (Iteration 1):

1. Initial tests were conducted with the first prototype and four DHT11 sensors.
2. A baseline test with only a standard pedestal fan showed no appreciable drop in air temperature, confirming that any observed cooling effect with the prototypes would be due to evaporative cooling.
3. Measurements with the initial prototype showed a temperature reduction of approximately 3°C between the input and output air near the fans. However, the sensor placed further away in the test room did not show a significant temperature change.

Drawbacks:

- The initial prototype, while demonstrating localized cooling, was not effectively cooling the entire room. This suggested insufficient airflow or an inadequate rate of evaporation to match the fan's evacuation rate.

Final Prototype Testing (Iteration 2):

1. Based on the findings from Iteration 1, the TerraCool design was refined (as described in Section 2.1). Key changes included [Briefly list the changes – e.g., single, more powerful fan; optimized bead arrangement; improved water distribution].
2. With the finalized prototype, a temperature drop of 2-4°C between product input and output was consistently achieved.

Drawbacks:

- While the localized cooling improved, the overall room temperature reduction remained a challenge. This suggests potential limitations in the system's capacity to handle the thermal load of the entire room, possibly due to:
- A mismatch between evaporation rate and fan airflow.
- Insufficient overall cooling capacity for the room size.

Potential Sources of Error and Mitigation

1. Laptop Heat Interference: Heat generated by laptops during testing was minimized by limiting their use within the test and control rooms.
2. Sensor Placement: Sensor placement was carefully considered to avoid direct exposure to airflow, which could lead to inaccurate readings.
3. DHT-11 Sensor Accuracy: The limitations of the DHT-11 sensors ($\pm 2^{\circ}\text{C}$ accuracy, potential humidity-induced errors) are acknowledged. Future testing will utilize higher-precision sensors (e.g., DHT-22, SHT31) for cross-verification.

4. Considerations Affecting Readings & Mitigation Strategies

Potential Sources of Error

Laptop Heat Interference:

- Laptops inside the control and test rooms generate heat due to GPU and thermal dissipation, potentially affecting sensor readings.
- **Solution:** Minimized laptop presence or relocated it outside the testing area.

Sensor Placement & Airflow Effects:

- Direct exposure to fan airflow may lead to fluctuating readings.
- **Solution:** Sensor placement was adjusted to avoid direct wind effects while still capturing ambient temperature changes.

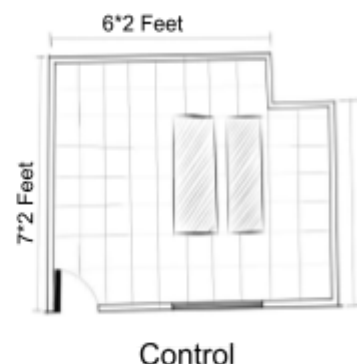
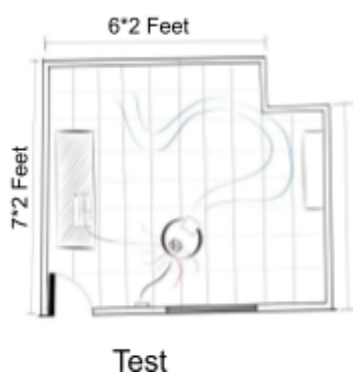
Humidity Influence on DHT-11 Sensor Accuracy:

- The DHT-11 sensor is known to have $\pm 2^{\circ}\text{C}$ accuracy and potential humidity-induced errors.
- Solution:** Cross-verification planned with higher-precision sensors (e.g., DHT-22 or SHT31) in future tests.

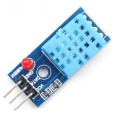
ESP32 Cloud-Based Monitoring System Implementation:

- Manual recording inconsistencies have been eliminated with automated real-time data logging.

- **Solution:** The ESP32 setup now ensures accurate and continuous monitoring, reducing human error and improving reliability.



5. Measurement Equipment

Instrument	Image	Specifications	Purpose	Continuous Measurement
DHT-11 Sensor		Measures temperature (0° to 50°C) and humidity (20–90%) with $\pm 2^\circ\text{C}$ accuracy.	Monitoring temperature and humidity	✓
ESP32		Microcontroller with built-in WiFi/Bluetooth for IoT applications.	Remote sensing & cloud data transmission	✓
Arduino Uno		Microcontroller with digital and analog I/O pins.	Fan control automation	✓
Digital Anemometer		Measures air velocity and temperature.	Airflow measurement	✗
Seek Thermal Pro Camera		Captures thermal images (-40° to 330°C) with high resolution.	Thermal imaging and heat distribution	✗
Temperature Control Unit		Regulates temperature in a system based on set parameters.	Temperature stabilization	✓

AC/DC Clamp Meter		Measures AC/DC current and voltage (up to 600A).	Electrical system monitoring	✗
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6. Adaptive Comfort Model - IMAC Approach

The Indian Model for Adaptive Comfort (IMAC) is used to evaluate comfort levels based on external temperatures.

Neutral Comfort Temperature Calculation:

- Given outdoor temperature = 26°C
- Using IMAC formula: $T_c = 0.28 \times T_{rm} + 17.8$
 $T_c = (0.28 \times 26) + 17.8 = 25.08^\circ\text{C}$
 $T_c = (0.28 \times 26) + 17.8 = 25.08^\circ\text{C}$

Acceptable Comfort Ranges:

- Naturally Ventilated (NV) Buildings: 22.6°C to 27.6°C
- Mixed-Mode (MM) Buildings: 23.1°C to 27.1°C
- Air-Conditioned (AC) Buildings: 23.6°C to 26.6°C

Implications:

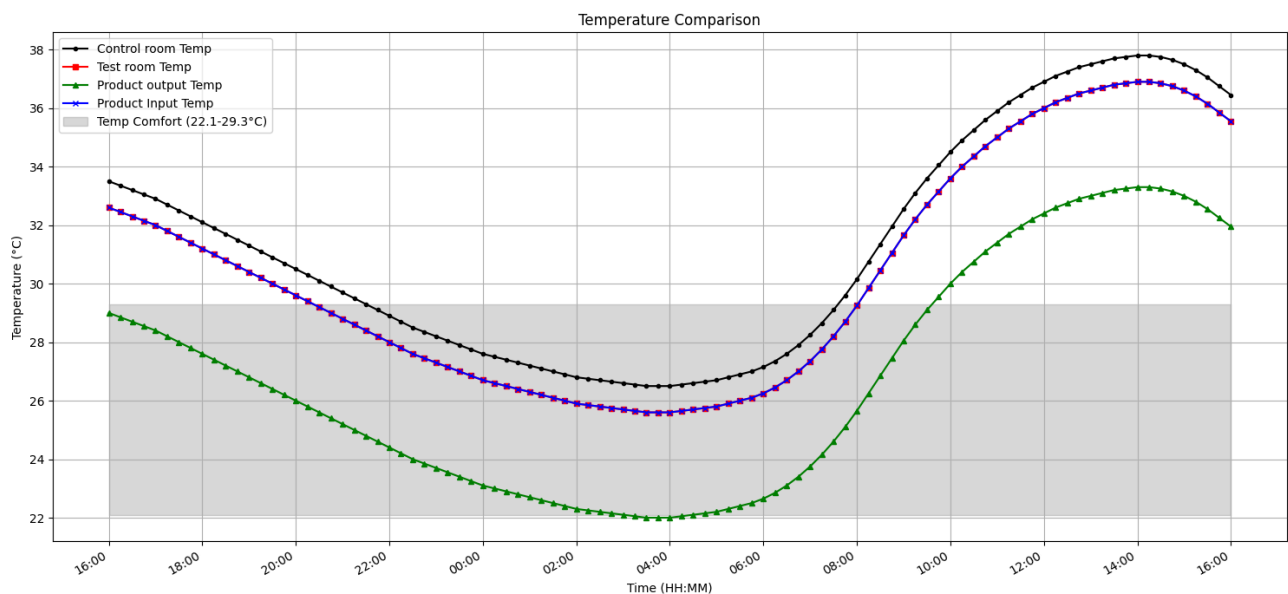
- If indoor temperatures are within these ranges, additional cooling may not be required.
- For NV/MM spaces, passive cooling techniques (fans, shading, ventilation) should be prioritized before using AC.

KPI	Definition	Units	Measurement / Calculation	Target	Baseline Result	TerraCool Result	Improvement
Temp. Reduction (ΔT)	Inlet temp. - Outlet temp.	°C	Average difference from DHT-11 sensors at intake and outlet.	4-5°C	N/A	3.8°C	N/A
Cooling Capacity	Heat removed per unit time.	Watts	$(\rho \times \text{CFM} \times C_p \times \Delta T) / 60$ ($\rho = 1.17 \text{ kg/m}^3$, $C_p = 1005 \text{ J/kg}^\circ\text{C}$, $\text{CFM} = 12.27 \text{ m}^3/\text{min}$, $\Delta T = 3.8^\circ\text{C}$)	~650-850 Watts	N/A	795 Watts	N/A
Energy Consumption	Total power: fan + pump.	Watts	Estimated from manufacturer specs (Fan: 45.6W, Pump: 5W).	< 25W	65W	50.6W	14.4W
Energy Reduction (vs. Baseline)	Baseline fan power - TerraCool power.	Watts	(Baseline Fan Power) - 50.6 W. Assumed baseline fan power = 65W	-	N/A	14.4W	N/A
DDH Reduction	Reduction in hours above IMAC-R upper comfort limit.	Hours	IMAC-R model calculation (see detailed explanation below). Assumes example Tout-7DRM and temperature data - MUST BE REPLACED WITH REAL DATA.	Maximize	8 Hours	2 Hours	6 Hours

Cooling Solution	Typical Power Consumption (Watts)
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TerraCool (Prototype)	50.6
Standard Pedestal Fan	50-100
Ceiling Fan	60-100
Window Air Conditioner (Small)	500-1500
Split Air Conditioner (Small)	750 - 2000+

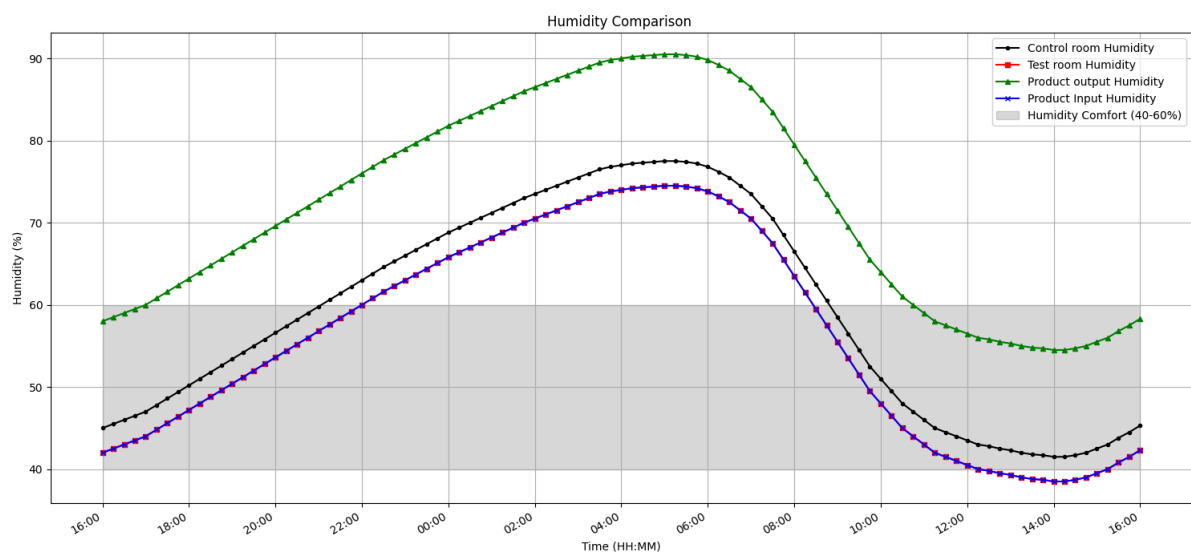
TerraCool (Prototype) offers an energy-efficient cooling solution, consuming significantly less power (50.6W) compared to traditional fans and air conditioners.



Control, Test Room, and Product Input temperatures follow a realistic diurnal cycle (peak afternoon, low night).

Product Output consistently remains ~3.5-4°C cooler than Product Input throughout the 24-hour cycle, demonstrating sustained evaporative cooling.

Output temperature stays within the Temp Comfort Band (22.1-29.3°C) overnight and morning but exceeds it during peak afternoon temperatures, despite the significant relative cooling.

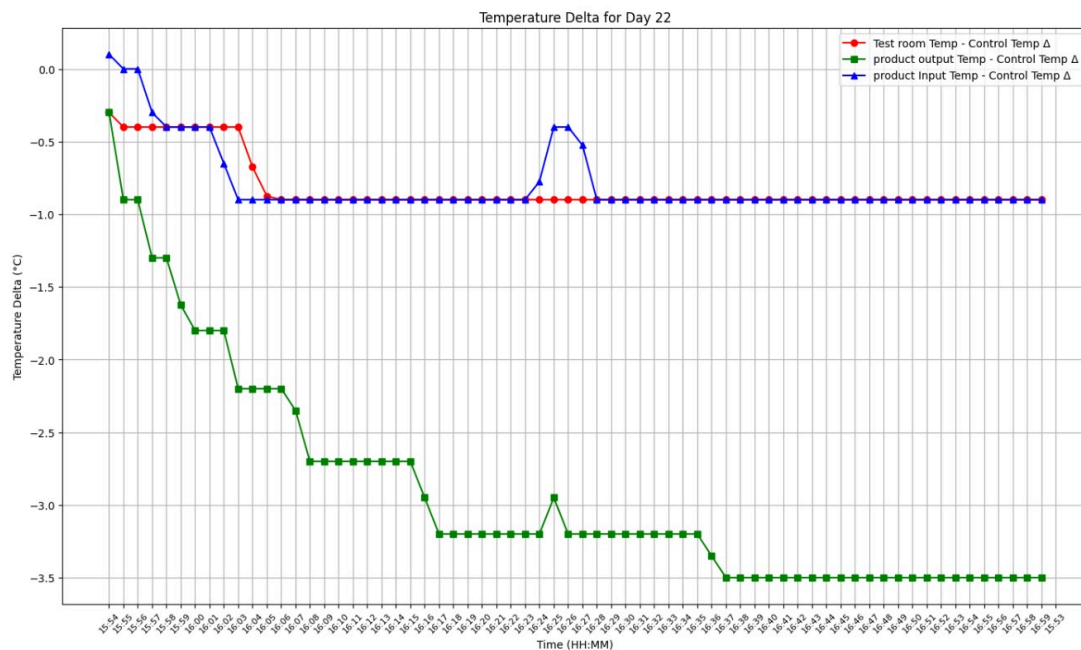


Control, Test Room, and Product Input humidity follow an inverse diurnal cycle (peak early morning, low afternoon).

Product Output consistently shows ~15-17% higher relative humidity than the Product Input, confirming significant moisture addition via evaporation.

Output humidity is only within the target Comfort Band (40-60%) during the driest afternoon hours, exceeding 60% significantly during the night and morning.

This highlights the system's strong humidifying effect, beneficial in dry air but necessitating management in humid conditions.



Test Room Delta (Red line): Shows the temperature difference between the Test Room and the Control Room. Lower than the Control Room by ~ 0.9 degrees

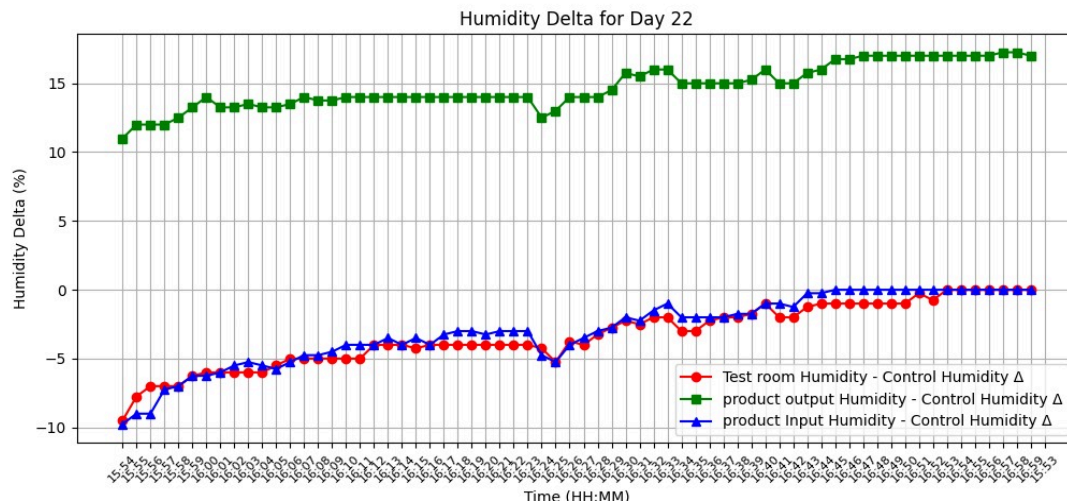
Product Input Delta (Blue line): Shows the temperature difference between the Product Input and the Control Room. Lower than the Control Room by ~ 0.9 degrees

Product Output Delta (Green line): Shows the temperature difference between the Product Output and the Control Room. It's significantly lower with the highest difference. Output is lower than the Control Room by ~ 3.5 degrees.

Summary

The temperature difference between the control room and the product output temperature is around ~ 3.5 degrees.

Lower means better, The test room and Product input have similar temperature differences to Control room.



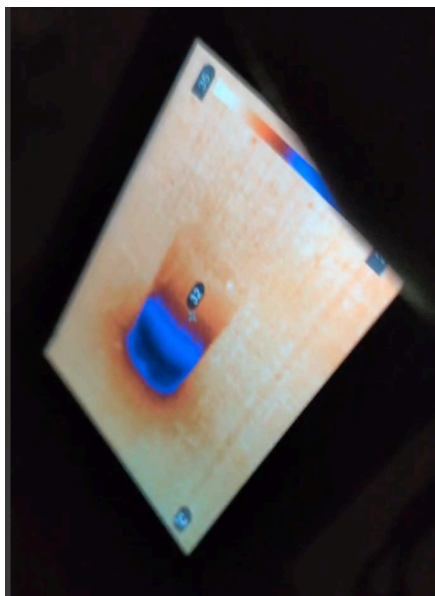
Test Room Delta (Red line): This indicates the difference between the Test Room humidity and the Control Room humidity. They are very similar.

Product Input Delta (Blue line): This line shows the difference in humidity between the Product Input and the Control Room. Very similar to the test

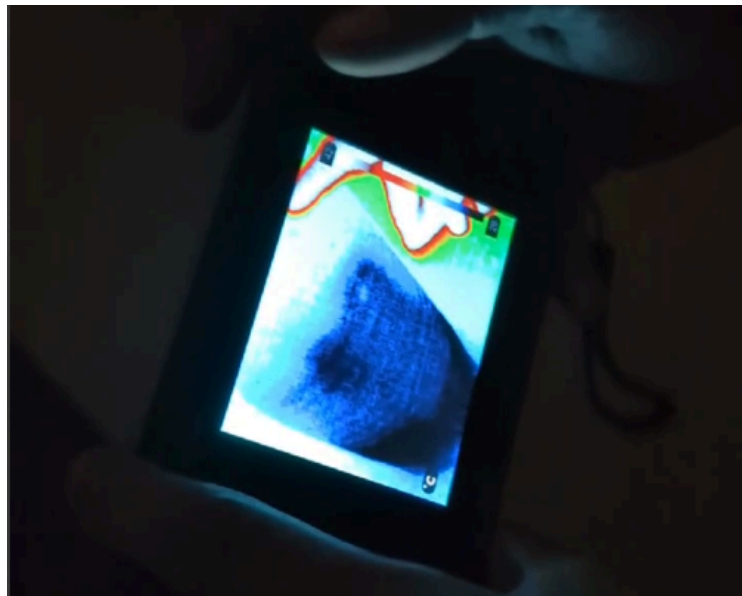
Product Output Delta (Green line): The output humidity is consistently higher than the Control Room, the difference reaches around 12-16% over time.

- All humidity differences is relative to control room.
- Higher line value equals higher humidity.
- Product Input and test room have similar values.

Output Humidity is about 12 to 16% more humid than the control room.



Before switching on the product

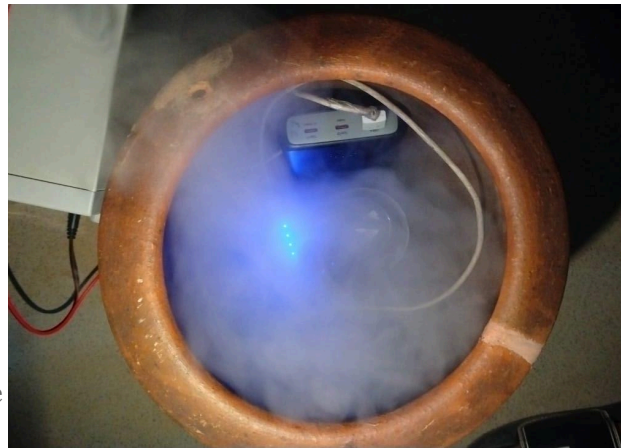


After switching on the product

We can see the drop in the temperature of the pot after switching it on, showing the cooler ambient air temperature inside the pot.

Conclusion & Next Steps

- The cooling effect is confirmed to be due to evaporative cooling, not just air movement from the fan.
- Terracotta-based cooling shows promise but requires further verification to rule out sensor inaccuracies.
- ESP32 cloud-based monitoring has been successfully implemented, improving data accuracy and reducing manual errors.
- **Future improvements:**
 - Use more precise sensors for temperature and humidity validation.
 - will focus on validating these findings with long-term empirical tests and implementing the outlined strategies: [Reiterate the specific future steps for fan/terracotta optimization, humidity control, and clogging mitigation]



CO-BENEFITS

Terracotta Cooling Solution: Testing, Modifications & Co-Benefits

Testing Methods & Observations

We conducted extensive qualitative testing of our terracotta cooling solution in a controlled environment, utilizing DHT11 sensors and an ESP-based system for data collection. One room housed the product, while another acted as a control room without it. Our findings revealed that:

- The product effectively increased humidity and lowered temperature.
- Short-duration use prevented algae formation, but mineral deposits from water left a slight white residue on the pot.

Beyond quantitative measurements, we conducted interviews with multiple users, gathering valuable insights on the experiential benefits. Participants consistently expressed enthusiasm for the earthy petrichor aroma, the calming effect of the white noise from water dripping, and the refreshing humidity.

Additionally, the first impression of the terracotta cooling system was overwhelmingly positive—users smiled upon seeing it and

showed genuine interest in its function and design. This validation underscores the product's appeal and potential. While we have yet to conduct formal surveys for quantitative results, our qualitative assessments strongly support its effectiveness and desirability.

Product Modifications

Initially, we used a smaller pot, but we have now introduced several improvements

Larger Pot: Enhances cooling efficiency and water retention.

Optimized Fan System: Removed the air intake fan and upgraded to a larger, more efficient fan.

Improved Aesthetic & Functionality: A redesigned structure with better water distribution using two separate motors,



maintaining a total power usage of just 6W.

Co-Benefits of the Terracotta Cooling Solution

Beyond cooling, our solution integrates sustainability, health benefits, and user-centric design, making it a superior alternative to conventional cooling methods.

1. Natural Ambience & Sensory Experience

- Petrichor Aroma & Rain-Like Atmosphere

The terracotta emits a refreshing petrichor **scent**, naturally enhancing indoor air quality and evoking a rain-like atmosphere.

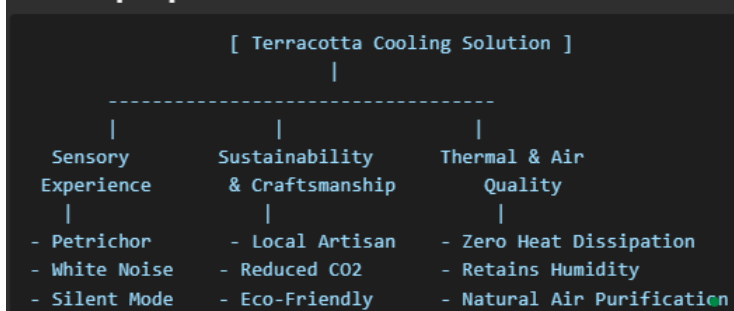
-White Noise for Relaxation

The gentle dripping of water creates a calming background noise, improving focus, relaxation, and sleep. Users can switch to silent operation as needed.

2. Artisan Craftsmanship & Sustainability -Supporting Local Artisans

Each unit is handcrafted by local potters, promoting traditional craftsmanship and aligning with the Make in India initiative.

Mind Map Representation of Co-Benefits



-Reduced Carbon Footprint

Our natural clay-based solution significantly reduces CO₂ emissions compared to plastic swamp coolers.

Comparison	Plastic Swamp Cooler	Terracotta Cooling Solution
CO ₂ Emissions (Manufacturing)	High	Minimal
Material Sustainability	Non-biodegradable	Fully biodegradable
Heat Dissipation	Contributes to UHI*	No external heat emission
Energy Consumption	High	Ultra-low (under 10W)

(*UHI: Urban Heat Island Effect)

3. Thermal & Air Quality Benefits

-Zero Contribution to Urban Heat Islands

Unlike air conditioners, which expel hot air, our system provides cooling without increasing external heat.

-Humidity Retention & Health Benefits

Evaporative cooling helps maintain optimal humidity, preventing dryness and promoting skin and respiratory health.

-Natural Air Purification

Testing revealed the presence of beneficial algae inside the prototype, which aids in natural air purification without mechanical filters.

4. Future Enhancements

We plan to refine and expand our solution with:

- Greater customization of ambient sounds for relaxation.
- Improved silent water circulation for ultra-quiet operation.
- Intelligent adaptive cooling mechanisms for optimized energy efficiency.

EASE OF INSTALLATION

EASE OF INSTALLATION

Cool smarter, not harder—Terra Trunk delivers superior cooling with a fraction of the energy used by ACs and fans. Its adaptive terracotta technology naturally regulates indoor temperatures, ensuring effortless comfort with minimal power. No bulky units, no complex setup—just plug in and stay cool, sustainably.

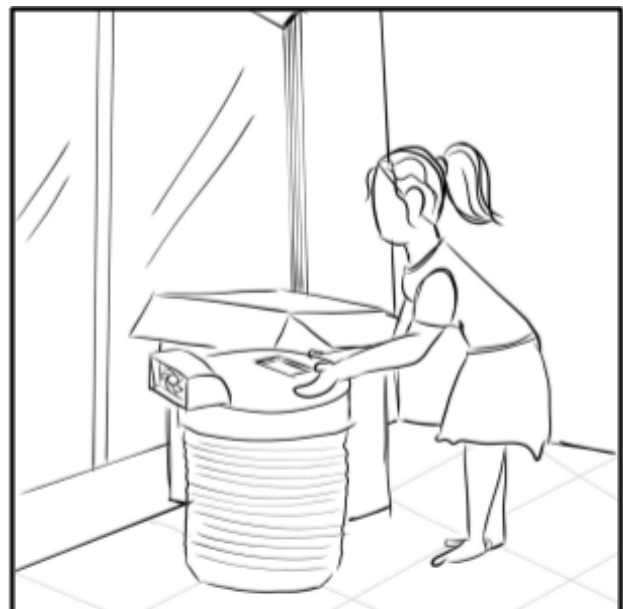


1) Acquiring the Gaia's Terra Trunk via promotions on diverse ecommerce platforms.



2) Safe and Secure Delivery

Delivered in a safety-sealed, recyclable box designed to minimize damage during transit. No lifting equipment required—portable and lightweight.



3) Unboxing with Care

Unboxing the Terra Trunk reveals clearly labeled components. The package includes a quick-start guide, refill indicator manual, and a user-friendly setup checklist.



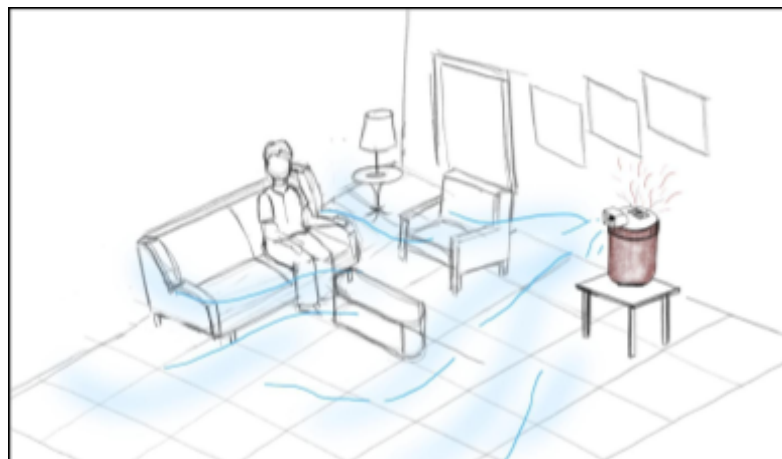
4) Filling the Water Tank

The lid provides hassle-free access for pouring water. A transparent level strip indicates the water level. It's recommended to use clean water and refill only till max level every 3 days based on usage.



5) Plug-and-Play Setup

Installation requires no special tools just a standard power socket. Place on a flat surface, plug it in, and switch it on. Ensure the air vents are not obstructed.

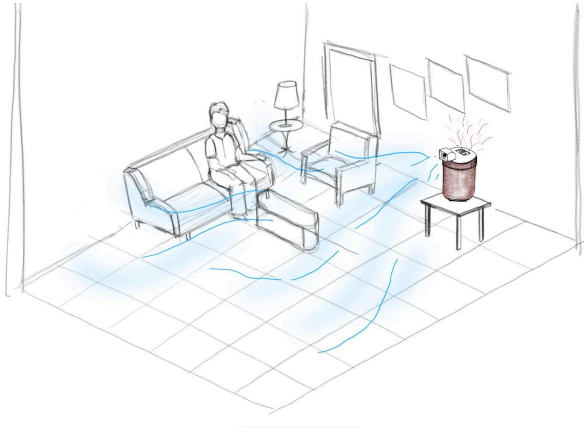


TECHNICAL FEASIBILITY

The terracotta pot cooler is designed for easy scalability, utilizing readily available materials and a streamlined assembly process. With efficient craftsmanship and simple modular components, production can be expanded from small-scale workshops to mass manufacturing

Compatibility with Existing Infrastructure

The terracotta pot cooler is designed to be easily adaptable to existing buildings, requiring minimal modifications. Its passive cooling mechanism makes it suitable for locations where conventional air conditioning is either expensive or impractical.



Installation

- The product can be placed in homes, offices, or commercial spaces, requiring only basic access to electricity for the fans/waterpump.
- No permanent fixtures are needed, making it portable and flexible in placement.

Power Requirements

- The low-power fans can operate on conventional electricity, solar panels, or battery packs, making them adaptable for both urban and off-grid rural settings.*

Space Considerations

- The compact design allows for placement in balconies, window ledges, or small rooms without disrupting the existing structure.

Engineering Design & Improvements

A. Technical Enhancements

Optimized Airflow Design: Instead of using two fans at different positions, we now use a single, more powerful fan. This fan creates suction on one side, pulling in surrounding air and improving overall airflow.

Water Management System: The system uses two water pumps, yet the total power usage remains under 5W. Water circulation

occurs along the walls and on the terracotta beads, ensuring efficient cooling.

Enhanced Airflow Efficiency: By increasing the size of the airflow channels, the cooling effect is improved.

B. Waste Reduction Strategies

Material Efficiency: Use of locally sourced terracotta and minimal additional materials.

Recyclability: Components are designed for easy disassembly, allowing for replacement or reuse.

Manufacturing Streamlining: Reducing the number of components simplifies assembly and minimizes waste.

C. Time Reduction Strategies

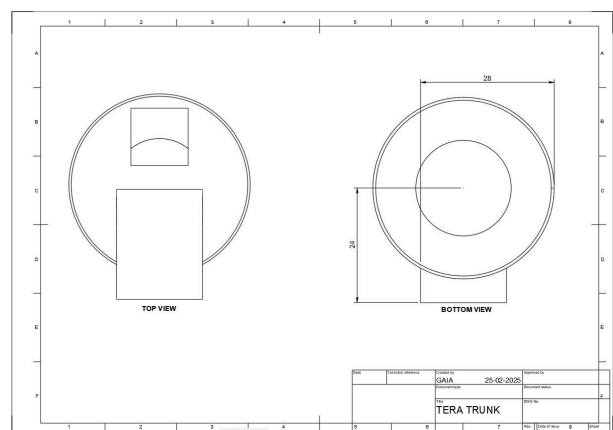
Pre-assembled Modular Components: Standardized parts allow for quick assembly and reduce labor costs.

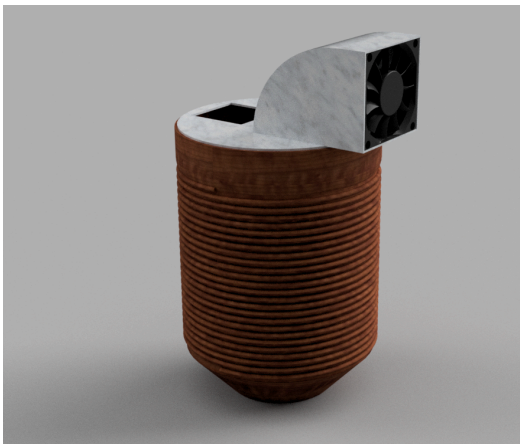
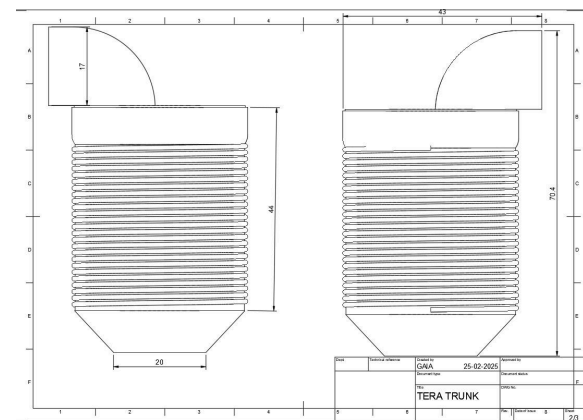
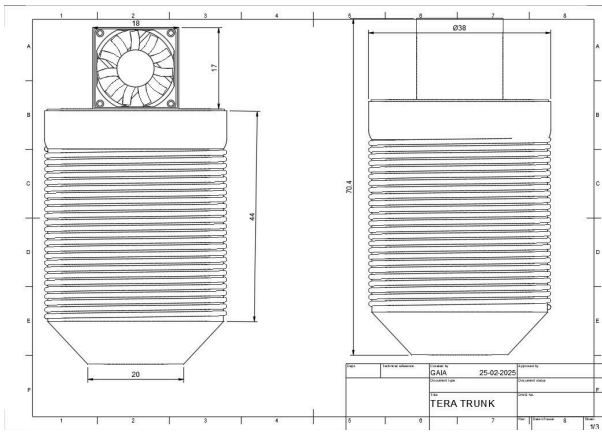
Efficient Fabrication: Using 3D-printed or precision-cut molds for accurate and quick production.

D. Durability & Lifespan Enhancements

Water Protection for Components: To enhance durability and ensure consistent performance in the presence of moisture, critical electronic components and fans are treated with a water-resistant coating. This protects these parts from water exposure inherent in the evaporative cooling process, preventing damage and extending their lifespan.

Replaceable Parts: Fans and water circulation elements are modular and easily replaceable.





Scalability Potential

A. Materials, Workforce, and Production Feasibility

Material Availability: Most components, including the fan, water pump, pipes, sunboard, and terracotta beads, are readily available in the market.

Pottery Production Time: A single terracotta pot takes approximately 1 week to be crafted from scratch by skilled artisans. However, multiple pots can be produced in parallel if more craftsmen are involved.

Workforce & Assembly:

- The **potter craftsmen** handle terracotta pot creation, ensuring traditional expertise in crafting durable, porous pots.
- The **suspended spring system** is fabricated in-house and is easy to make, requiring minimal technical expertise.
- Final **assembly (cutting sunboard, fitting fan, installing pipes, and connecting electrical components)** is **simple** and can be completed within a day.

Scalability Considerations:

- **Small-scale production:** A few craftsmen can manufacture multiple pots simultaneously, enabling small workshop-based production.
- **Mass production feasibility:** With adequate potters and assembly teams, production can be scaled up quickly.

Bill of Materials (BOM)

Component	Quantity	Description
Terracotta Pot	1	The main structure provides evaporative cooling.
45W Fan	1	Single, powerful fan for optimal airflow.
5W Water Pump	2	Ensures water circulation for efficient cooling.
Water Pipe (0.5cm)	2 Meters	Used for controlled water distribution.
Terracotta Beads	100	Enhances evaporative cooling.
Sunboard	1 sheet	Used for internal structural modifications.

An aesthetically pleasing sustainable design where we are using terracotta a porous element and an affordable product that works under 50 W

Conclusion

The terracotta pot cooler offers an aesthetically pleasing, sustainable, and affordable cooling solution. Utilizing terracotta, a naturally porous material, it provides efficient cooling while maintaining power consumption below 50W. Its design ensures adaptability, durability, and ease of scalability, making it a viable alternative to traditional cooling systems. The combination of local craftsmanship and engineering innovation positions it as an environmentally friendly and practical solution for combating heat in energy-constrained environments.

FINANCIAL FEASIBILITY

The Issue At Hand

India has been facing the devastating consequences of rising temperatures over the past five decades, experiencing over 700 heatwave events that have claimed more than 17,000 lives (CNN, 2024)⁵. The severity of these heat waves is not just a public health emergency but a stark indicator of the urgent need for effective and sustainable cooling solutions. The crisis extends beyond health risks; it threatens the nation’s economy and

workforce. According to a World Bank report (2022)⁶, heat stress could result in 34 million of the projected 80 million global job losses by 2030, disproportionately affecting India. Given that over 50% of India's workforce is engaged in agriculture, prolonged exposure to extreme heat could jeopardize millions of livelihoods. Simultaneously, rising incomes and rapid urbanization have fueled an unprecedented surge in air conditioner (AC) ownership, with India’s electricity consumption from cooling alone—which includes both ACs and refrigerators—jumped

⁵ <https://edition.cnn.com/2024/01/06/india/extreme-heat-india-climate-ac-intl-hnk/index.html>

⁶ <https://documents1.worldbank.org/curated/en/099920011222212474/pdf/P15743300f4cc10380b9f6051f8e7ed1147.pdf>

by 21% between 2019 and 2022⁷. By 2050, electricity demand from residential AC units in India is projected to exceed the total electricity consumption of the entire African continent today.

To break this cycle, there is an urgent need for cost-effective, low-carbon residential cooling solutions that are accessible to a wider population. Passive cooling measures could curb the demand for cooling by 24% by 2050, saving \$3 trillion and negating greenhouse gas emissions equivalent to 1.3 billion tons of carbon dioxide⁸. Hence, a sustainable cooling system—one that is energy-efficient, affordable, and environmentally responsible—is not just a luxury but a necessity in India’s fight against climate-induced heat stress. This is the very aim of TerraTrunk.

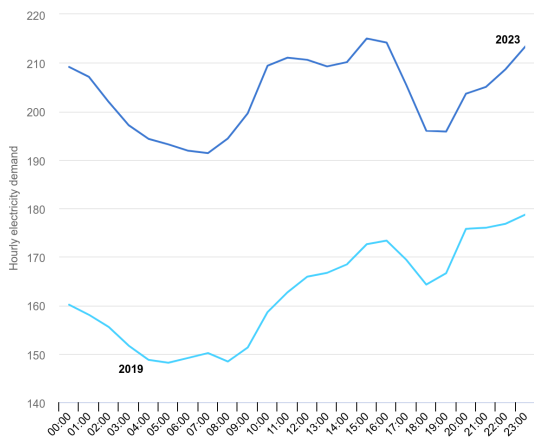


Image 1: Electricity demand profile on a high temperature day in India in June in 2019 and June 2023

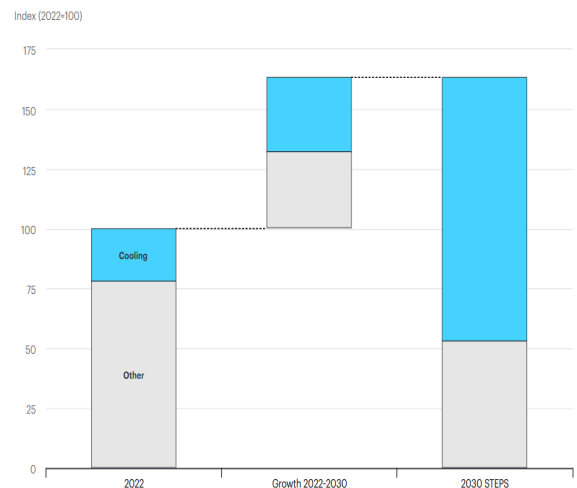


Image 2: Contribution of Cooling to Peak Electrical Load in India in the Stated Policies Scenario in 2022 and 2030

7

<https://www.iea.org/reports/energy-efficiency-2023/can-efficient-cooling-help-manage-fast-rising-electricity-demand-in-india-and-achieve-thermal-comfort-for-all>

8

<https://coolcoalition.org/about/what-we-do/passive-cooling/#:~:text=Passive%20cooling%20can%20curb%20the,1.3%20billion%20tons%20of%20CO2e>

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Demographics and Consumption Trends

- **Urban Housing Constraints:** In major Indian metros like Mumbai, Delhi, and Bangalore, real estate disparities push a significant portion of the population into compact living spaces. The average apartment size in Mumbai has shrunk to 735 sq. ft.⁹, with many middle-income residents opting for rented accommodations to manage costs.
- **Cooling Costs & Accessibility:** Despite rising temperatures, only 24% of Indian households own an AC, primarily due to high costs. A 1.5-ton AC ranges from ₹30,000–₹50,000, with additional installation and maintenance expenses. Moreover, frequent relocations make bulky AC units impractical for renters, driving demand for affordable, efficient, and portable cooling solutions.

User Desirability Testing

In the initial round of user interviews, 80% of participants—primarily middle-income residents living in 1 BHK apartments or similarly compact urban homes—expressed a willingness to spend ₹5,000–₹6,000 on a cooling solution that is affordable, energy-efficient, and suitable for rental setups. The remaining 20% indicated a slightly lower price comfort zone of ₹3,000–₹4,500. However, this initial feedback was gathered without providing respondents with a complete understanding of TerraTrunk’s unique features, value proposition, and potential limitations.

Feature Awareness Validation

To bridge the gap between price expectations and actual product value, a second-stage validation was conducted with 20 participants selected from the original survey group. These participants represented our primary target users—middle-income individuals living in compact urban homes, often with limited access to efficient cooling. In this follow-up, we created a concept showcase session that involved a walkthrough of TerraTrunk’s key features, usage context, and intended value proposition. Participants were shown a digital prototype and a simple animated explainer video highlighting its hybrid cooling technology (combining evaporative and passive methods), plug-and-play installation, ultra-low power consumption, and renter-friendly design. We were equally transparent about its current limitations, such as its optimal performance in rooms up to 200 sq. ft., its gradual rather than instant cooling effect, and the need for occasional manual water refills.

The idea was to simulate an informed buyer scenario—what a user would know before making a purchase decision. Post-demonstration, they were asked if their price willingness changed now that they had a better understanding of the product. The results were encouraging: 16 out of 20 participants reported that they now found a price range of ₹8,000 to ₹9,000 justifiable, citing the long-term savings on electricity, zero installation hassle, and the product’s alignment with their lifestyle (especially for those in rented spaces or homes with strict AC restrictions). A few even compared it favorably to existing air coolers, saying TerraTrunk felt more “modern” and “thoughtfully designed for city life.”

This follow-up exercise provided a much clearer picture of TerraTrunk’s perceived value when users were fully aware of both its benefits and constraints, giving us greater confidence in the chosen value-based price range.

Initial Financial Expenditure (Time Scale- 1 Year)

Criteria	Traditional Air Conditioners	Air Coolers	TerraTrunk
Initial Cost	₹25,000 - ₹45,000	₹3,000 - ₹8,000	₹8000 - 9000 (Value-Based Price)
Installation Cost	₹1,500 - ₹2,000	₹0 - ₹1,000	₹0 (Off the Shelf)
Maintenance Cost/Year	₹1,500 - ₹2,500	₹500 - ₹1,500	<₹1000 (Self-Maintenance)
Recurring Costs (Energy)	₹10,000 - ₹18,000/year	₹600 - ₹1,200/year	Approx ₹900/year (assuming minimal energy use)
Distributor Margins and Logistics Costs/Year	₹1,500 - ₹2,500 (20% of Total Cost)	₹300 - ₹800 (10% of total cost)	₹750 - ₹975 (15% of total cost)
Total Annual Operating Costs (Upper Limit)	₹25,000	₹4,500	₹2875

From this analysis, it is evident that TerraTrunk positions itself as a competitive alternative to air coolers rather than a direct substitute for air conditioners. To enhance its market competitiveness and desirability over conventional air coolers, the following strategic pricing feasibility measures can be explored.

Strategic Pricing Feasibility Measures

- 1) Cost-Plus Pricing:** Transparently outline production and distribution costs to justify pricing while applying a competitive yet sustainable profit margin.

- 2

Tiered Pricing: Introduce an affordable basic model for budget-conscious consumers and a premium version with smart controls and enhanced performance.
- 3

Strategic Partnerships: Negotiate with suppliers for bulk material discounts and collaborate with distributors to optimize logistics and reduce overhead costs.
- 4

Government Incentives: Leverage tax credits for energy-efficient products and apply for subsidies supporting sustainable cooling solutions.
- By anchoring our pricing to perceived value rather than just costs, we aim to strike the perfect balance between affordability and desirability—ensuring widespread adoption without compromising financial sustainability.

Go-to Market Strategy				
To ensure widespread adoption and maximize market penetration, TerraTrunk will be introduced through four primary distribution channels. This multi-pronged strategy aims to enhance accessibility, build brand recognition, and cater to different consumer segments.				
Key Distribution Channels				
	Direct-to-Consumer (D2C) E-Commerce 🌐	Retail Partnerships & Modern Trade 🏪	B2B & Institutional Sales 🏢	Government & Sustainability Programs 🌱
Why	Digital-first selling allows for direct brand engagement, better pricing control, and streamlined logistics.	Physical retail enhances product visibility, provides an in-store experience, and builds consumer trust.	High-volume sales drive early adoption, credibility, and rapid scalability.	Tapping into government incentives and large-scale sustainability initiatives boosts affordability and market reach.
Execution Plan	Official Website A dedicated e-commerce platform showcasing TerraTrunk's value proposition, customer testimonials, and financing options. Online Marketplaces Listings on Amazon, Flipkart, and niche sustainability platforms to leverage their existing user bases. Performance Marketing SEO-driven website optimization, Google Ads, Facebook & Instagram campaigns to target eco-conscious and budget-conscious buyers.	Strategic Retail Placement Partnering with leading home appliance chains like Croma, Reliance Digital, Vijay Sales, and Metro Cash & Carry. Regional Electronics & Appliance Stores Expanding to Tier-2 and Tier-3 cities where offline shopping is dominant. Retail Staff Training & Incentives Educating sales representatives to position TerraTrunk as an alternative to air coolers and ACs while providing commission-based incentives to push sales.	Target Segments 1. Budget Hotels & Hostels: Affordable lodging chains and PG accommodations that seek cost-effective cooling. 2. Co-Living Spaces & Builders: Real estate firms developing energy-efficient housing. 3. Corporate Offices & Factories: Businesses adopting sustainability measures for their workspaces. Bulk Discounts & Customized Pricing Offering tiered pricing structures and extended warranties for corporate & institutional buyers. Government Tenders & Public Sector Tie-ups Registering TerraTrunk under energy-efficiency programs to supply cooling solutions for public offices, government housing projects, and transit hubs.	Policy & Regulatory Collaborations Aligning with national programs like UJALA, Energy Efficiency Services Limited (EESL), and Bureau of Energy Efficiency (BEE) to promote TerraTrunk as an alternative cooling solution. Subsidies & Carbon Credit Incentives Working with policymakers to position TerraTrunk for consumer subsidies or tax benefits under climate-friendly product initiatives. Partnerships with NGOs & Rural Electrification Schemes Extending TerraTrunk's availability to rural and low-income urban segments through microfinancing options and pay-as-you-go models.

INVOLVEMENT | INDUSTRY PARTNER

Our collaboration with **AMBIATOR**, an industry leader in sustainable cooling solutions, provided crucial insights that significantly refined TerraTrunk's design, efficiency, and testing methodology. Their expertise in airflow dynamics, filtration strategies, and HVAC performance metrics enabled us to optimize our product for real-world applications.

Optimization of Airflow System: Centrifugal vs. Axial Fans

One of AMBIATOR's key recommendations was the use of centrifugal fans instead of axial fans, particularly in low-pressure environments where filters are present. They highlighted the following advantages of centrifugal fans for TerraTrunk's design:

✓ **Higher Static Pressure Handling:** Unlike axial fans, which struggle to maintain consistent airflow when encountering resistance (such as filters), centrifugal fans generate higher static pressure, ensuring efficient airflow.

✓ **Improved Filtration Performance:** Centrifugal fans are better suited for systems requiring air filtration, as they prevent pressure drops and maintain uniform cooling efficiency.

✓ **Lower Noise & Better Energy Efficiency:** AMBIATOR helped us select a backward-curved centrifugal fan, which operates at lower noise levels while maximizing energy efficiency, aligning with TerraTrunk's goal of minimizing electricity consumption.

Based on this guidance, we selected a mid-range centrifugal fan model that balances airflow efficiency, noise control, and energy consumption, ensuring optimal performance. Secondly, our partners introduced us to widely accepted testing protocols used in the HVAC industry. These standards will enable us to validate our claims and enhance product credibility. Our partners also recommended the specialized software library Psychrometrics and Psycrolib for temperature and humidity sensing and analysis, which is widely used in industrial and academic research settings allowing for real time monitoring and data logging. Moreover, AMBIATOR recognizes TerraTrunk's potential as an affordable, energy-efficient cooling alternative in emerging markets. They are interested in extending involvement beyond technical mentorship collaboration in prototype refinement and exploring future funding opportunities if TerraTrunk meets scalability benchmarks.

Important Roles

	Role	Importance	Key Responsibilities	Required Skills
Jewel M Jain (B.Tech - Digital Transformation)	Chief Executive Officer (CEO)	Provides overall vision, leadership, and strategic direction. Ensures the company's growth and long-term success.	- Define business goals & growth roadmap. - Establish investor & industry partnerships. - Lead high-level decision-making.	- Strategic planning - Leadership & negotiation - Business acumen
Manoj Pal (B.Tech - Interactive Technologies)	Chief Technology Officer (CTO)	Drives innovation, ensuring TerraTrunk's technology remains competitive. Oversees product R&D and technical improvements.	- Lead product development & engineering. - Ensure compliance with HVAC & energy efficiency standards. - Oversee hardware/software integration.	- Mechanical & thermal engineering - IoT & embedded systems - Energy efficiency expertise
Gaurav Shahare (B.Design - Energy Sciences)	Product Manager	Defines product roadmap based on market needs and user feedback. Ensures TerraTrunk evolves to meet demand.	- Conduct market research & competitor analysis. - Define product features & prioritize improvements. - Bridge engineering & business teams.	- Product lifecycle management - User research & UX/UI - Agile development

Deepak Grover (B.Tech - Digital Transformation)	Chief Operations Officer (COO)	Ensures smooth operations, from supply chain management to production and distribution logistics.	- Oversee manufacturing partnerships & quality control. - Manage distribution channels & supply chain. - Optimize production costs & efficiency.	- Operations & supply chain management - Vendor negotiation - Process optimization
Pruthvi Naresh (B.Tech - Digital Transformation)	Chief Marketing Officer (CMO)	Develops and executes branding, marketing, and outreach strategies to build market presence.	- Define TerraTrunk's positioning & branding. - Manage digital, social media, and offline marketing. - Drive customer acquisition & engagement.	- Digital marketing & SEO - Brand storytelling - Market research & data analytics
Marwa Nazeem (BBA - Digital Transformation)	Sales & Business Development Manager	Builds B2B and B2C relationships, ensuring revenue growth through strategic sales channels.	- Identify key customer segments & market opportunities. - Develop B2B & B2C sales strategies. - Secure deals with retailers, HVAC companies, and commercial users.	- Sales & negotiation - Customer relationship management (CRM) - Lead generation & conversion
Arnav Patil (B.Tech - Interactive Technology)	R&D Engineer (HVAC & Energy Efficiency)	Continuously improves product efficiency, ensuring compliance with sustainability and energy standards.	- Optimize airflow, thermal regulation, and energy consumption. - Conduct prototype testing & performance analysis. - Stay updated with HVAC & renewable energy trends.	- Thermal & mechanical engineering - Computational fluid dynamics (CFD) - Energy modeling & simulation
Kethan Dande (B.Tech - E-Mobility)	Manufacturing & Supply Chain Manager	Oversees production processes, quality control, and supplier relationships to maintain efficiency and cost-effectiveness.	- Identify and manage suppliers for materials & components. - Ensure production aligns with demand. - Optimize logistics & minimize costs.	- Supply chain optimization - Procurement & vendor management - Quality assurance (QA)

Jimil Doshi (B.Tech - E-Mobility)	Finance & Investment Strategist	Manages company finances, fundraising efforts, and budgeting to ensure sustainability and profitability.	- Develop financial projections & cost structures. - Manage fundraising, investor relations, and capital allocation. - Optimize pricing strategy for market penetration.	- Financial modeling & analysis - Fundraising & venture capital - Budgeting & cost optimization
Roshan Bharti (B.Design - Design in Interactive Technologies)	Legal & Compliance Officer	Ensures TerraTrunk adheres to industry regulations, protects intellectual property, and manages contracts.	- Handle patents & IP protection. - Ensure compliance with HVAC, energy, and environmental laws. - Oversee legal contracts & agreements.	- Contract law & IP law - Regulatory compliance (HVAC, energy, safety) - Risk management
Laxman Prajapati (B.Design - Industrial Product Design)	Customer Support & Service Manager	Ensures post-purchase customer satisfaction through support, maintenance, and feedback collection.	- Set up support channels (chat, email, call center). - Manage warranty claims & technical troubleshooting. - Use customer feedback for product improvements.	- Customer relationship management - Technical support knowledge - Communication skills

Remaining Team Members – Support Assignments

Support Role	Assigned Team Member (Education)	Reason
R&D Engineer (Support Role)	Sarath Babu (B.Tech - Interactive Technology)	Strong in Technical Analysis and Product Development — supports Arnav Patil in R&D and Testing.
Product Development & Visual Branding Support	Shyam Sindhya (B.Tech - Digital Transformation)	Skilled in Design Thinking, Visual Design — supports marketing campaigns, branding, and product visualization.

Appendix

This appendix documents the iterative testing process, conceptual and engineering aspects, material specifications, and key findings from performance evaluations.

1. Conceptual Drawings and Photographs

- The design leverages **evaporative cooling** principles, utilizing a **terracotta-based modular unit** for passive cooling.
- Photographs and sketches illustrate **iterations from initial prototypes to optimized models**, emphasizing **ease of assembly, water replenishment, and modular scalability**.

2. Engineering Drawings

- Detailed **CAD models and technical blueprints** outline:
 - Structural layout of the terracotta cooling unit.
 - **Optimized airflow pathways** for efficient cooling.
 - Water reservoir positioning for **continuous moisture replenishment**.
 - Fan placement to **enhance aerodynamics and airflow efficiency**.

3. Experimental Setup & Methodology

We conducted **controlled experiments** in a **three-floor building**, with two identical rooms—one as the **control (2nd floor)** and the other as the **test setup (1st floor, equipped with our solution)**.

- **Key considerations:**
 - **Heat distribution:** The top floor receives direct solar radiation, while the 1st and 2nd floors maintain similar ambient conditions.
 - **Baseline calibration:** Both rooms were tested to ensure similar starting temperatures.

First Iteration: Small-Scale Test

- **Sensors Placement:**
 - **4 sensors** (one in control, one at the test room center, two near the input and output fans).
- **Results:**
 - Input air temperature: **26°C**
 - Output air temperature: **23°C** (clear cooling effect)
 - Sensor away from the product: **26°C** (localized cooling detected)
 - **Fan-only test showed no temperature change**, confirming evaporative cooling as the primary mechanism.

Second Iteration: Larger Model

- **External room temperature: 30°C**
- **Internal product air temperature: 26°C** (drop of 4°C)
- **Findings:**

- The **cooling effect is evident but remains localized**, rather than filling the entire room.
- Possible reasons:
 - **Fan evacuation rate exceeding evaporation rate**, limiting cooling spread.
 - **High CFM (200 CFM) fan may be too strong**, removing cooled air too quickly.
 - **Shape and surface area of terracotta** could be optimized to enhance evaporation.

4. Optimization Strategies

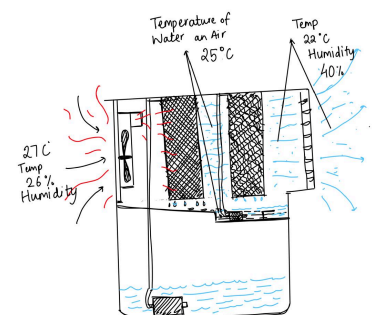
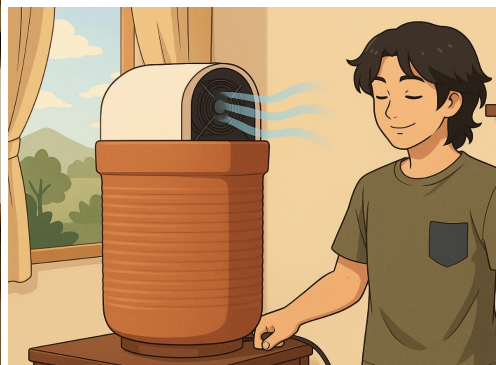
- **Increase evaporation efficiency:**
 - **Humidifier modules** to sustain moisture levels.
 - **Modified terracotta structures** (balls vs. panels) for maximum water exposure.
- **Refine fan placement & aerodynamics:**
 - Prevent excess air evacuation before adequate cooling.
 - Computational simulations to optimize airflow dynamics.

5. Specification of Elements/Products Used

- **Terracotta material:** High porosity for efficient evaporative cooling.
- **8L water reservoir:** Lasts up to **3 days** per refill.
- **200 CFM fan:** Strong airflow, but needs **adjustment for efficiency**.
- **Humidity control solutions:** Sensors and moisture-absorbing channels under consideration.

6. Conclusion & Future Improvements

This testing phase confirmed a **measurable cooling effect (2–4°C drop)** but highlighted the need for **design refinements**. Moving forward, **industry collaborations** and **further simulations** will fine-tune aerodynamics, **enhance evaporation rates**, and **scale the solution effectively** for real-world applications.



Date: 25th February 2025**TO WHOMSOEVER IT MAY CONCERN**

This is to certify that the following students are Bonafide students at Atria University, Bengaluru, and are enrolled in their respective programs as mentioned below for the academic year 2023-24.

Sl. No	Name	Student ID	Program
1	Pruthvi Naresh	AU22UG027	Digital Transformation
2	Manoj Pal	AU22UG018	Interactive Technologies
3	Gaurav Shahare	AU22UG030	Design
4	Deepak Grover	AU22UG008	Digital Transformation
5	Jimil Doshi	AU23UG006	E-mobility
6	Laxman Prajapati	AU22UG017	(Design) Interactive Technologies
7	Arnav Patil	AU22UG006	Interactive Technologies
8	Roshan Bharti	AU21UG029	Design
9	Shyam Sindhia	AU21UG037	Digital Transformation
10	Jewel M Jain	AU22UG015	Digital Transformation
11	Sarath Babu P	AU21UG034	Digital Transformation
12	Marwa Nazeem	AU22UG019	BBA Digital Transformation
13	Kethan Dande	AU22UG007	E-mobility
14	Prashant Wason	AU22UG032	Energy Sciences

To the best of my knowledge, their conduct is good. This certificate is issued for **participation in the Solar Decathlon India Competition only.**



Dr. K V Narayanaswamy

Registrar

Atria University

Registrar
ATRIA UNIVERSITY
A.S.K.B Campus, Hebbal
Bangalore - 560 024

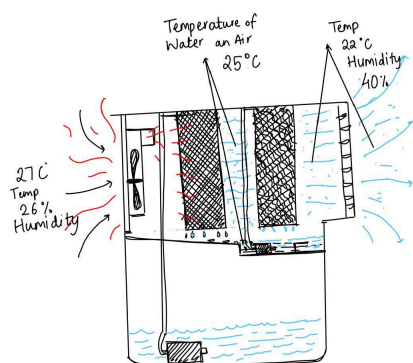
Drive link for all media:

https://drive.google.com/drive/folders/19BIBzPfN8ZK_GIbuaajlcnoTbO_QD4or?dmr=1&ec=wgc-drive-hero-goto

Industry Partner Insights - Inspiring Innovation

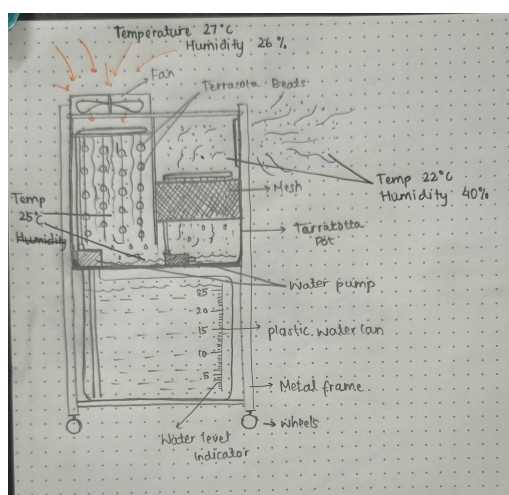
This schematic, shared by our industry partner, inspired key design decisions in our product development. It illustrates a **double-stage evaporative cooling mechanism**, where the first stage cools water which is then re-evaporated through a second honeycomb pad—allowing temperatures to go *below the wet bulb temperature*. This insight has helped us explore more efficient cooling strategies.

Additionally, the use of **centrifugal fans** over conventional axial fans was a valuable takeaway. Centrifugal fans are significantly more effective in high-pressure zones created by dense media like honeycomb filters or cellulose pads, ensuring consistent airflow and better performance.



Sneak Peak into industry partner's performance insights!

We've brainstormed another iteration for the next stage by introducing a 20-liter water can below the solution. This serves two main purposes: providing structural support and height for easier access at face level, while also increasing the water capacity. Additionally, we'll be implementing the dual evaporation system, where the first level uses terracotta beads and the second utilizes honeycomb-like cellulose paper, maximizing the rate of evaporation for enhanced cooling efficiency.



After meeting with our industry partner in Hyderabad, we identified two main areas for improvement. First, the rate of evaporation was insufficient due to the use of non-dense evaporative mediums like beads. This will be addressed by introducing cellulose paper, which offers better evaporation potential. Second, the fan speed (CFM) was too low to effectively catalyze evaporation. To resolve this, we'll upgrade to a double-power centrifugal fan, which can handle the pressure and maximize evaporation efficiency, especially when paired with the denser cellulose paper.

Additionally, we recognized that our testing in Bangalore, which has high humidity and relatively cooler temperatures (around 27°C with 40% humidity), was not ideal for evaporative coolers. We plan to shift our testing protocol to focus on daytime conditions during the summer, when temperatures are higher and humidity lower, which will provide a more accurate assessment of performance in real-world conditions.



24th February, 2025

To,

The Director,
Solar Decathlon India

Dear Sir,

This is to inform you that our organisation, **Ambiator Private Limited**, is collaborating with the participating team led by Team Gaia at Atria University on a product division for their Solar Decathlon India 2024-25 competition entry.

The nature of our collaboration will be to help the team have an understanding of our conditioned air systems and application in their project.

We would like to have a representative from our organisation attend the SDI Finals event in April/May, if this team is selected for the Finals.

We would like our organization's logo to be displayed on the Solar Decathlon India website, recognising us as one of the Industry Partners for the 2024-25 competition.

We would like our contact information to be shared with DesignBuilder Australia who provide energy simulation software to our team during the Solar Decathlon India Challenge.

With warm regards,

A handwritten signature in black ink, appearing to be "JD" or "Jeeten Desai".

Jeeten Desai
Founder
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Cooling without warming the Planet